

Kernel-Native Upstream Continuation and Projection Intertwining under AASC

A Fixed-Clock RPI Semiflow Theorem with Lawful Metric-Orbit Finality,
Clock-Conditioned Full-Ray Quantum Descent, Einstein and Schrödinger
Intertwining, and Distinct Closed/Exchange-Extended Cubical Coherence

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Capstone gate addressed: G5 - upstream continuation and projection commutation

Abstract

This manuscript repairs and completes the continuation object required by the AASC gravity–quantum capstone. The dependency order is immutable: target adequacy forces the AASC kernel; the kernel governs the realized physical interior; gravity and quantum standing are downstream roles of that interior; Einstein and Schrödinger dynamics are later bookkeeping projections; accepted Modules A and B are later still. No clock, seed, metric development, Hilbert flow, source tensor, Hamiltonian, or exchange ledger authorizes an upstream role.

The construction fixes one certified local physical-clock representative and forms a clock-normalized, locally Lipschitz seed directly on a regular chart of the lawful RPI quotient. Standard local-flow theory constructs the upstream semiflow before projection. Clock-forward Moore segments form a fixed-clock history category, and an orientation-preserving change of clock representative induces an explicit duration-transport equivalence between separately constructed presentation categories.

The metric fibre is corrected to be lawful-redescription invariant: its complete profile is the groupoid orbit of the whole natural metric/source test functional, with metric, source, tests, boundary data, Ward status, and Cauchy data transported together. Equality of orbit profiles is proved equivalent to lawful metric/source equivalence. The quantum fibre is fixed by the full transition profile, operator graph/domain, spatial/source ledger, and full-ray process; record and compatibility use their imported final quotients.

The regular metric branch descends through a smooth local gauge slice and, on the Cauchy-eligible subtype, intertwines with the independently fixed Einstein–Module-A development. The quantum branch uses exact strongly continuous full-ray descent rather than treating an unbounded Stone generator as an upstream Banach vector field. Wigner, Stone, and the accepted Schrödinger normal-form theorem then classify the downstream quantum process. Record and compatibility descend by exact quotient preservation.

Open or effective continuation is also retyped. The closed prediction (Π_X^0, β_X^0) and the actual exchange-extended projection $(\tilde{\Pi}_X, \tilde{\beta}_X)$ are distinct maps and processes. A common upstream incidence ledger advances only the enriched role coordinates. Directed exchange cells therefore relate independently typed predicted and actual endpoints and satisfy identity and source-target-correct cocycle laws. The four role cells generate a genuine sixteen-vertex product cube whose faces and monotone paths commute; zero incidence gives the strict closed cube.

A complete closed inhabitant uses the globally smooth upstream seed $\partial_\theta + \partial_r$ on a clock-ray-record RPI packet. The full oscillator Stone flow appears only in the clock-conditioned quantum projection $\Pi_H(\theta, q, r) = R_\theta q$, which is surjective on the full ray carrier and intertwines exactly with R_t . A second packet adds one smooth common-ledger coordinate and gives two composable nonzero exchange segments without altering the closed metric process. A bounded checker validates only the finite registry of state samples, separate base and enriched endpoints, incidence restrictions, all sixteen vertices, and composition cocycles; it does not claim to verify the analytic theorems.

Dependency lock.

$$\text{TargetAdeq}_D \implies K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D) \implies A_D^+ \implies I_D.$$

Only then:

$$\begin{aligned} I_D &\implies \{\text{Gravity}_D, \text{QuantumStand}_D\}, \\ \{\text{Gravity}_D, \text{QuantumStand}_D\} &\implies \{\text{Einstein}_D, \text{Schrödinger}_D\}, \\ \{\text{Einstein}_D, \text{Schrödinger}_D\} &\implies \text{accepted Modules A–B} \implies \text{Module C}. \end{aligned}$$

The continuation seed is an explicit tensor occupant inside the already kernel-governed interior. It does not generate the kernel or the interior from below.

Reportable endpoint. In the declared regular local fixed-clock envelope, an independently formed smooth RPI seed generates a nonempty clock-forward Moore-history category. Four independently complete role fibres carry unique descended metric, quantum, record, and compatibility processes. The lawful metric orbit intertwines with the Einstein–Module-A Cauchy development on its exact eligible subtype; the full-ray quantum branch is clock-conditioned downstream and is Wigner–Stone–Schrödinger classified without an upstream unbounded-generator field. Closed role projections form a strict four-cube. Distinct closed-prediction and exchange-extended actual projections form a coherent nonzero exchange cube generated by one common incidence ledger. Explicit closed and declared-open nonidentity RPI packets inhabit every mandatory certificate coordinate.

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Part I

Dependency Lock and Standalone Module Target

1 Proof class and exact standalone target

Accepted Modules A and B close source-representation, finite composite-source, common-charge, and stable scalar proportionality burdens in their exact envelopes. Neither module constructs a physical history. Module C addresses only the continuation problem: form one standing-bearing lineage at the RPI level and then prove that the metric, quantum, record, and compatibility histories are faithful process projections of that lineage.

The proof class is *fixed-clock kernel-native continuation construction followed by independently proved quotient-process descent*. It is not a pullback of solved developments, not metric-first Cauchy construction, not Hilbert-first unitary postulation, and not a claim that the kernel selects a preferred dynamics.

Definition 1.1 (Module C target class). A *Module C target* is a fixed regular local continuation envelope containing:

1. a local quotient chart U_D of the realized physical interior $\bar{I}_D = I_D / \sim_I$ and its complete fixed RPI role packet;
2. one certified local physical-clock representative $\tau : U_D \rightarrow J$ for one declared use;
3. a standing-bearing local seed V_D on U_D , supplied as target tensor content rather than generated by the kernel;
4. standing, failure-boundary, record-lineage, response, and declared-exchange data;
5. candidate metric, quantum, record, and compatibility probe families whose completeness is not assumed and must be proved in Part IV;
6. the accepted Module A source interface on the source-bearing metric subtype and accepted Module B only where its stable scalar envelope is invoked.

The target does not contain downstream Einstein or Schrödinger histories, fibre-completeness conclusions, the exchange cube, or the commutation theorem.

Proposition 1.2 (No downstream pullback as the upstream object). *A pullback of solved metric, Hilbert, record, and compatibility developments may certify agreement of those developments over a common base, but cannot establish their descent from an independently formed RPI continuation.*

Proof. Such a pullback exists only after its downstream legs exist, and its commutative squares are membership conditions. Module C has the reverse order: V_D and its flow are formed on U_D first;

four role fibres are then proved complete; their quotient processes are then descended and classified. Compatibility by selection is not upstream construction. $\square$

Nonclaim 1.3 (No dynamics from the bare kernel). The theorem does not assert

$$K_D \implies V_D.$$

The kernel governs any determinate continuation. The seed is the explicit continuation-role occupant supplied by the fixed realization certificate.

The endpoint is local and comparison-class relative. It does not prove global clock or flow completeness, nonzero GR/QM interaction, arbitrary distributional Einstein well-posedness, or the capstone unique assignment orbit.

2 Imported theorem-use packages

Every imported result is used only at the role stated here. The local continuation theorem is new to this manuscript; the imported papers do not already supply it.

2.1 Realized physical interior

The RPI theorem constructs the role architecture

$$I_D = (\text{NonVacRec}, [B]_{\equiv_B}, \text{LCD}, \text{Cmp}, \text{BoundTrans}, \Phi, \Pi, G, M, \Omega, T, \text{Cau})$$

and proves coherence, role closure, and operational minimality. Its continuation, time, record, and causal roles are upstream of every metric or Hilbert representation [1]. This paper uses RPI to fix the state, tensor, record, transport, and failure signatures on which the continuation seed acts. It does not infer RPI from the success of the seed.

2.2 Coherent trajectories

The bivalent trajectory theory supplies legal trajectories, coherence as standing at every stage, prefix closure, composition closure, identity-preserving invariance, tensor determination, no deferred coherence, and irrecoverability of a failed prefix [2]. Those results are applied after the local flow is constructed, to certify its finite samplings and its Moore-segment category.

2.3 Physical clock quotient

The clock theorem identifies a physical clock with the use-relative quotient of a complete certified clock-use profile, not with a preferred variable or external time parameter [3]. Module C fixes one local certified class and one representative readout. A different representative gives a separately constructed Moore category related by duration transport, not an internal duration quotient. The clock theorem is not used to create the kernel or the continuation seed, and no global preferred clock is claimed.

2.4 Connection and persistence

The standing-preserving connection theorem supplies quotient-stable transport comparison along admitted continuation morphisms, identity normalization, restriction, and weak composition compatibility [4]. The persistence-realization theorem says that an already existing irreversible standing-bearing continuation induces a persistence-regime equivalence class; it does not generate that continuation [5]. Module C uses these packages after the upstream history category exists.

2.5 Einstein, Schrödinger, records, and compatibility

The Einstein theorem identifies the minimal first-response metric equation in the faithful metric envelope with a closed or declared-exchange source [6]. The Schrödinger theorem identifies the minimal closed autonomous scalar nonrelativistic Hamiltonian flow after projective standing, Wigner implementation, strong continuity, Stone generation, and the spatial first-response predicate are established [7]. The record theorem supplies the measurement-record quotient and fixation discipline [8]; the entanglement theorem supplies compatibility-not-transport discipline [9]. Structural Unification supplies the required ledger coordinates and warns against projection backflow [10].

2.6 Accepted Modules A and B

Accepted Module A constructs

$$q_g : \widehat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}$$

from attachment-certified upstream metric-response witnesses and proves Ward/divergence and finite-composite closure [11]. Accepted Module B constructs the stable scalar common-charge factorization and calibrated active/inertial equality in its exact matching envelope [12]. Module C uses neither result to form V_D ; they enter only when the metric and stable scalar projection fibres are evaluated.

Import boundary. A theorem-use citation is not a substitute for a missing construction. In particular, coherent trajectory theory presupposes a licensed transition relation, the clock theorem presupposes clock candidates and a declared use, and persistence realization presupposes irreversible standing-bearing continuation. Module C constructs the missing local flow and upstream history category explicitly.

3 Frozen standalone endpoint and proof gates

The frozen endpoint is unchanged. The corrections below alter neither its strength nor its dependency order; they make the clock, fibre, process, exchange, cube, inhabitation, and software types exact.

Definition 3.1 (Module C certificate). A complete Module C certificate is a tuple

$$\mathcal{C}_D^C = (C1, \dots, C12)$$

with:

C1	fixed kernel-governed RPI quotient chart, complete RPI role packet, standing/failure loci, and target identity;
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<i>C2</i>	one certified local physical-clock representative, oriented readout, declared use, and failure ledger;
<i>C3</i>	explicit locally Lipschitz upstream continuation seed V_D , nonzero on at least one certified state and formed before projection;
<i>C4</i>	standing tangency, irreversible record order, complete path-sample registration, and no-repair data;
<i>C5</i>	local-flow existence/uniqueness and a fixed-clock Moore-history category; clock-equivalent representatives are related only by an external duration-transporting equivalence;
<i>C6</i>	independently proved complete metric, quantum, record, and compatibility fibres;
<i>C7</i>	smooth closed evolution for the regular metric profile, exact strongly continuous full-ray evolution for the quantum profile, and exact fibre preservation for record/compatibility profiles;
<i>C8</i>	accepted Module A source ancestry, Einstein normal-form eligibility, and—where dynamical intertwining is claimed—a registered smooth hyperbolic Cauchy system and uniqueness certificate;
<i>C9</i>	full-ray saturation, transition preservation, strong continuity, domain, Stone, and Schrödinger spatial eligibility on the quantum branch;
<i>C10</i>	record-fixation and compatibility-not-transport quotient certificates;
<i>C11</i>	a common exchange-incidence monoid, role restriction actions, identity/cocycle laws, and the resulting strict or exchange-certified four-cube;
<i>C12</i>	at least one complete nonidentity upstream inhabitant and a bounded finite audit certificate.

Definition 3.2 (Standalone proof gates). The module closes only if all of the following are proved.

- C1.** The continuation seed is formed on the upstream RPI quotient before any downstream process.
- C2.** The seed generates a local flow and a nonempty forward fixed-clock Moore-history category.
- C3.** Standing and failure status are preserved and no failed prefix is repaired.
- C4.** Each projection fibre is proved complete by an independent separation or final-quotient theorem.
- C5.** Closed profile evolution implies unique descended metric, quantum, record, and compatibility processes without assigning smooth structure to discrete quotient coordinates.
- C6.** The smooth metric Cauchy-eligible process is the unique local Einstein–Module-A development of its projected initial data; the weaker distributional branch is reported only as pointwise normal-form classification.
- C7.** The quantum process lies in the Wigner–Stone–Schrödinger class on its eligible subtype.
- C8.** Record and compatibility processes preserve their final quotient profiles and do not launder update, source transfer, or transport.
- C9.** Open/effective discrepancies are restrictions of one common directed incidence and satisfy identity and cocycle laws in explicit action categories.

C10. The sixteen vertices and every face/cycle of the four-role continuation cube commute strictly or through the declared common exchange.

C11. At least one nonidentity upstream morphism inhabits every mandatory certificate coordinate.

Nonclaim 3.3 (Excluded promotions). The module does not report: dynamics generated from the kernel; a global or preferred clock; all Einstein–matter Cauchy problems; nonzero interacting quantum gravity; universal source attachment for every quantum role; Module D fibre uniqueness; the capstone unique orbit; or strong equivalence.

Part II

Clock-Conditioned Upstream Continuation

4 The upstream RPI quotient chart

Definition 4.1 (Physical quotient chart). Let X_D^+ be the standing state family of the fixed realized physical interior and let $\sim_I$ be lawful RPI redescription, equality of the complete RPI role profile supplied by the RPI theorem. A *regular local physical chart* is a quotient submersion

$$q_I : X_D^+ \supset \tilde{U}_D \twoheadrightarrow U_D \subset \bar{I}_D$$

onto a C^1 Banach manifold on the declared regular stratum. The chart carries the fixed role tuple

$$\mathfrak{I}_D = (\text{NonVacRec}, B, \text{LCD}, \text{Cmp}, \text{BoundTrans}, \Phi, \Pi, G, M, \Omega, T, \text{Cau})_D$$

from the realized-physical-interior theorem, together with its record, failure, boundary, and exchange refinements.

The Banach chart is downstream local representation, not AMetric primitive structure. A chart change preserving every RPI role coordinate is skin. A change of domain, boundary, differentiability, response carrier, continuation law, or standing classification changes the theorem object.

Definition 4.2 (Standing and failure loci). The chart contains an embedded standing submanifold $\mathcal{S}_D \subseteq U_D$ and a disjoint registered failure locus $\mathcal{F}_D$. A state of $\mathcal{S}_D$ carries the full fixed RPI target identity. A point of $\mathcal{F}_D$ is a failure witness and cannot be reclassified as standing in the same retained history.

Definition 4.3 (RPI role signature). For $x \in U_D$, write

$$\Sigma_I(x) = (\Sigma_{\text{loc}}, \Sigma_{\text{trans}}, \Sigma_{\text{field}}, \Sigma_{\text{part}}, \Sigma_{\text{gauge}}, \Sigma_{\text{resp}}, \Sigma_{\text{vac}}, \Sigma_{\text{clock}}, \Sigma_{\text{rec}}, \Sigma_{\text{comp}}, \Sigma_{\text{fail}}, \Sigma_{\text{exch}})(x).$$

This is the fixed RPI role signature. Its role-level completeness is the imported RPI coherence/closure/minimality result: deleting or changing a displayed role changes the realized-physical-

interior target. It is not yet a claim that the four later projection profiles are complete; those four separation theorems are proved independently in Part IV.

Proposition 4.4 (No downstream solution in chart formation). *Neither an Einstein development nor a Hilbert-space orbit is required to define U_D , $\mathcal{S}_D$, $\mathcal{F}_D$, or Σ_I .*

Proof. The objects are the RPI quotient state, its standing/failure classification, and its role-bearing locality, transport, carrier, persistence, response, vacuum, clock, record, compatibility, and exchange data. Metric and Hilbert representations are added only through later probe maps. Deleting those maps leaves the upstream chart and seed domain intact. $\square$

5 Certified local clock representative and presentation changes

A clock orders a declared continuation use; it does not select the continuation law. To avoid mixing distinct Moore categories, the theorem fixes one certified local representative throughout the construction.

Definition 5.1 (Certified fixed clock representative). Fix a declared clock use U , a nonempty certified clock class $[\Theta]_{\text{clk},U}$, and one representative

$$\tau : U_D \longrightarrow J \subseteq \mathbb{R}.$$

The representative carries the complete clock-use certificate: realized carrier, definedness domain, quotient-stable readout, retained record, response to admissible continuation, orientation, boundary/failure set, and lawful reuse for U . Turning points, periodic recurrences, and global extension beyond J are outside this local theorem object.

Proposition 5.2 (Clock representative is not the seed). *The fixed readout τ does not determine the continuation seed V_D .*

Proof. The clock certificate determines whether a readout has standing for the declared use. Distinct standing-bearing vector fields may satisfy the same clock normalization and record order. Therefore τ is an anchor/record coordinate and V_D is a separate tensor coordinate. $\square$

Definition 5.3 (Clock normalization). The seed is normalized to the fixed representative when

$$\text{d}\tau_x(V_D(x)) = 1$$

throughout its declared domain.

Definition 5.4 (Equivalent clock presentation). Let $\tau' = f \circ \tau$ for an orientation-preserving C^1 diffeomorphism f of the local readout interval that preserves the complete clock-use profile. Define the corresponding seed

$$V'_D = (f' \circ \tau)^{-1} V_D.$$

The pairs (τ, V_D) and (τ', V'_D) are distinct presentations of the same oriented unparameterized RPI histories.

Definition 5.5 (Duration transport). For a τ -Moore segment (x, T) , define

$$\rho_f(x, T) = f(\tau(x) + T) - f(\tau(x)).$$

This is the τ' -duration of the same geometric orbit segment.

Remark 5.6 (Type discipline). A nonlinear clock change does not act as an automorphism of one fixed Moore category. It will induce a functor between the separately constructed categories for (τ, V_D) and (τ', V'_D) . Module C never quotients those categories as though their duration coordinates were identical.

6 The standing-bearing continuation seed

Definition 6.1 (Upstream continuation seed). A *fixed-clock RPI continuation seed* on U_D is a vector field

$$V_D \in \Gamma^1(TU_D)$$

with the following certificate.

1. V_D is locally Lipschitz in every admitted chart and satisfies $d\tau(V_D) = 1$.
2. $V_D(x) \in T_x \mathcal{S}_D$ for $x \in \mathcal{S}_D$.
3. Every failure margin and no-repair marker is forward fail-closed.
4. The seed is specified on the full RPI role chart before the metric, Hilbert, record, or compatibility quotient process is constructed.
5. Smooth metric and quantum profile derivatives, where used, are registered as functions of the corresponding profile only.
6. Record and compatibility profiles satisfy the exact fibre-preservation certificates of Part IV; no tangent space is presumed for their discrete or quotient-valued coordinates.
7. Internal and external exchange coordinates are explicit RPI tensor data.
8. There exists $x_0 \in \mathcal{S}_D$ with $V_D(x_0) \neq 0$.

Remark 6.2 (Supplying the seed is not backflow). The kernel does not generate V_D . The seed is a declared continuation-role occupant in an already admitted RPI target. The proof burden is to construct its flow and establish its quotient and projection properties. A solved Einstein or Schrödinger path is not permitted as the definition of V_D .

Definition 6.3 (Smooth profile closure). For a differentiable profile map $P : U_D \rightarrow Y$ into a Banach manifold, the seed is *smoothly P-closed* when there exists a locally Lipschitz vector field F_P on $P(U_D)$ such that

$$dP_x(V_D(x)) = F_P(P(x)).$$

Definition 6.4 (Exact fibre closure). For an arbitrary profile map $P : U_D \rightarrow Y$, with no smooth structure assumed on Y , the seed is *exactly P-closed* when

$$P(x) = P(y) \implies P(\alpha_{t,s}x) = P(\alpha_{t,s}y)$$

for every common defined interval. This is the closure notion used for record and compatibility quotients.

7 Local flow construction

Theorem 7.1 (Fixed-clock upstream flow). *Let $\mathfrak{C}_D$ satisfy definition 6.1. For every $x \in \mathcal{S}_D$ there is a maximal open interval $I_x \ni 0$ and a unique C^1 integral curve*

$$\gamma_x : I_x \longrightarrow U_D, \quad \dot{\gamma}_x(t) = V_D(\gamma_x(t)), \quad \gamma_x(0) = x.$$

The curves assemble into a local flow

$$\alpha_{t,s}(x) = \gamma_x(t - s)$$

where defined, satisfying identity and cocycle laws. Moreover

$$\tau(\alpha_{t,s}x) = \tau(x) + (t - s).$$

Proof. In an admitted Banach chart, V_D is locally Lipschitz. Picard–Lindelöf theory gives the unique maximal integral curve. Uniqueness gives

$$\alpha_{t,u} \circ \alpha_{u,s} = \alpha_{t,s}$$

whenever both sides are defined. The fixed-clock normalization gives

$$\frac{d}{dt} \tau(\gamma_x(t)) = 1,$$

so integration yields the clock equation. □

Corollary 7.2 (Local invertibility versus admitted orientation). *The analytic local flow is invertible where both time directions remain in its domain. The admitted continuation category nevertheless contains only clock-forward segments. Mathematical invertibility of a chart does not erase retained record order or permit same-act repair of a failed history.*

Proposition 7.3 (Nonvacuity criterion). *If $V_D(x_0) \neq 0$, then for sufficiently small $T > 0$,*

$$\alpha_{T,0}(x_0) \neq x_0.$$

Proof. If $\alpha_{T,0}(x_0) = x_0$ for all sufficiently small T , differentiation at 0 would give $V_D(x_0) = 0$. Hence a nonzero seed produces a nonidentity local segment. □

Theorem 7.4 (Clock-presentation time change). *Let (τ', V'_D) be the equivalent clock presentation of definition 5.4. Then the same unparameterized geometric integral curves are obtained after the time change*

$$T' = \rho_f(x, T) = f(\tau(x) + T) - f(\tau(x)).$$

Proof. Along a V_D -curve, $\tau(t) = \tau(x) + t$. Writing $t' = f(\tau(x) + t) - f(\tau(x))$ gives

$$\frac{dt'}{dt} = f'(\tau(x) + t) > 0.$$

The chain rule then yields

$$\frac{d\gamma}{dt'} = (f' \circ \tau)^{-1} V_D = V'_D.$$

Thus the two presentations share the same oriented geometric paths while carrying transported durations. $\square$

8 Standing invariance, failure, and no repair

Theorem 8.1 (Standing preservation of the upstream flow). *Every orbit segment that remains in the declared local-flow domain and begins in $\mathcal{S}_D$ remains in $\mathcal{S}_D$.*

Proof. The seed is tangent to the embedded standing submanifold. The restricted vector field therefore has a local flow on $\mathcal{S}_D$. Inclusion into U_D gives an ambient integral curve with the same initial value, and uniqueness identifies the two flows. $\square$

Definition 8.2 (Admitted and failed segment). A clock-forward segment is *admitted* when every point of the segment lies in $\mathcal{S}_D$, every required role certificate remains defined, and every external exchange is declared. The first point at which one of these conditions fails is a failure witness. A failed segment is not a morphism of the admitted continuation category.

Definition 8.3 (Complete finite path sample). A bounded software certificate may register a finite ordered sample

$$x_0, x_1, \dots, x_n$$

of an analytic segment. It is *complete for the finite claim* when every sample used by the encoded composition, face, or exchange assertion appears in the registry. The software does not claim that a finite sample proves analytic standing at unregistered times; the manuscript theorem supplies that through theorem 8.1.

Theorem 8.4 (No deferred repair). *If a finite ordered sampling of a prospective segment has an incoherent prefix, no later extension of the same lineage is an admitted coherent history.*

Proof. Adjacent samples are joined by licensed subsegments of the upstream flow. The imported no-deferred-coherence and irrecoverability theorems apply. A later corrected construction may restart with a new target or certificate but cannot convert the failed prefix into the same admissible act. $\square$

Corollary 8.5 (Skin cannot repair a history). *Coordinate, gauge, basis, phase, notation, or report changes preserving the complete tensor signature cannot convert a failed segment into an admitted one.*

Part III

Moore History Category and Persistence

9 The fixed-clock Moore history category

Definition 9.1 (Moore history segment). A τ -Moore RPI history segment is a pair

$$(x, T), \quad x \in \mathcal{S}_D, \quad T \geq 0,$$

such that $\alpha_{t,0}(x)$ is defined and admitted for every $0 \leq t \leq T$. Its source and target are

$$s(x, T) = x, \quad t(x, T) = \alpha_{T,0}(x).$$

The duration belongs to the fixed clock presentation. It is retained so that concatenation is strict. A different clock representative produces a different Moore category connected later by duration transport.

Definition 9.2 (Raw fixed-clock Moore category). The category $\mathbf{Cont}_D^{\tau, \text{raw}}$ has objects the states of $\mathcal{S}_D$ and morphisms the admitted τ -Moore segments. For composable segments (x, T) and (y, S) with $y = \alpha_{T,0}(x)$, define

$$(y, S) \circ (x, T) = (x, T + S).$$

The identity at x is $(x, 0)$.

Lemma 9.3 (Strict associativity). *Composition in $\mathbf{Cont}_D^{\tau, \text{raw}}$ is strictly associative.*

Proof. Both bracketings of three composable segments produce $(x, T + S + R)$. The flow cocycle fixes the common endpoint. $\square$

Proposition 9.4 (Coherent finite samples). *Every ordered finite sample of a morphism of $\mathbf{Cont}_D^{\tau, \text{raw}}$ is a coherent legal trajectory.*

Proof. Every adjacent pair is joined by an admitted subsegment. Every sample lies in the standing locus by definition of a morphism. The imported trajectory characterization therefore gives coherence. $\square$

10 Lawful fixed-clock equivalence and clock-presentation functors

Definition 10.1 (Fixed-clock lawful history automorphism). A lawful fixed-clock history automorphism is a diffeomorphism $g : U_D \rightarrow U_D$ satisfying

$$\tau \circ g = \tau, \quad dg(V_D) = V_D \circ g,$$

and preserving the complete RPI role signature, standing/failure loci, boundary data, and exchange ledger. It sends every admitted τ -Moore segment (x, T) to $(g(x), T)$ and therefore acts as an endofunctor of $\mathbf{Cont}_D^{\tau, \text{raw}}$.

Definition 10.2 (Fixed-clock history congruence). The relation $\sim_{\text{hist}, \tau}$ is the smallest categorical congruence containing all fixed-clock lawful history automorphisms. Thus it preserves sources, targets, duration, composition, standing, failure, role profiles, and declared exchange.

Lemma 10.3 (Fixed-clock congruence exists). *The relation $\sim_{\text{hist}, \tau}$ exists and is compatible with Moore composition.*

Proof. Intersect all categorical congruences containing the fixed-clock lawful generators. The indiscrete congruence is one such relation, so the family is nonempty. Every generator is an endofunctor of one fixed raw category and preserves durations and all standing-bearing coordinates; the intersection is therefore a well-typed categorical congruence. $\square$

Definition 10.4 (Upstream fixed-clock continuation category). The upstream continuation category is

$$\mathbf{Cont}_D^\tau = \mathbf{Cont}_D^{\tau, \text{raw}} / \sim_{\text{hist}, \tau}.$$

Theorem 10.5 (Clock-presentation equivalence). *For an equivalent clock presentation (τ', V'_D) , the duration transport of definition 5.5 defines a functor*

$$\mathfrak{T}_f : \mathbf{Cont}_D^\tau \longrightarrow \mathbf{Cont}_D^{\tau'}, \quad [(x, T)] \longmapsto [(x, \rho_f(x, T))].$$

It is an equivalence with inverse $\mathfrak{T}_{f^{-1}}$ and preserves oriented unparameterized histories, endpoint order, standing, role profiles, failure, and exchange incidence.

Proof. By theorem 7.4, the transported segment has the same geometric path and endpoint. For composable segments,

$$\rho_f(x, T + S) = \rho_f(x, T) + \rho_f(\alpha_{T,0}x, S),$$

so $\mathfrak{T}_f$ preserves composition. The inverse time change gives the inverse functor. All non-duration role coordinates are unchanged. $\square$

Remark 10.6. Clock-equivalent presentations are therefore equivalent categories, not equal morphisms inside one category. The fixed-clock construction is sufficient for the standalone endpoint and does not privilege a global clock beyond its declared local use.

11 Upstream category theorem

Theorem 11.1 (Kernel-native fixed-clock continuation category). *A complete Module C seed constructs a nonempty category $\mathbf{Cont}_D^\tau$ with the following properties.*

1. *Objects and morphisms are formed from the RPI quotient, one certified local clock representative, and the upstream seed before projection.*
2. *Prefix restriction and endpoint-compatible composition preserve coherence.*
3. *Fixed-clock lawful redescription acts trivially on standing-bearing history content.*

4. Any failed prefix excludes every extension of that lineage.
5. If the seed is nonzero somewhere, $\mathbf{Cont}_D^\tau$ contains a nonidentity morphism.
6. Any equivalent clock representative yields a separately constructed but duration-transport equivalent category.

Proof. The flow theorem constructs $\mathbf{Cont}_D^{\tau, \text{raw}}$. Moore associativity and identity segments give a category. Standing preservation and proposition 9.4 give coherent finite restrictions. The fixed-clock congruence is well typed by lemma 10.3, so the quotient category is defined. No deferred repair follows from theorem 8.4. Nonidentity follows from proposition 7.3. Clock-presentation equivalence is theorem 10.5. $\square$

Corollary 11.2 (Upstream formation precedes projection). *Deleting every metric, quantum, record, and compatibility projection map leaves the construction of $\mathbf{Cont}_D^\tau$ intact.*

Proof. No downstream projection occurs in the definitions or theorems constructing the RPI chart, fixed clock, seed, flow, Moore category, or fixed-clock quotient. $\square$

12 Persistence-regime status of the upstream lineage

Definition 12.1 (Irreversible history certificate). An upstream history has the irreversible certificate when its retained-consequence profile is nondecreasing, at least one admitted segment changes that profile, and no clock-forward extension can erase the change while preserving the same lineage.

Theorem 12.2 (Persistence-regime induction). *Every nondegenerate Module C history carrying the irreversible certificate induces a persistence-regime equivalence class*

$$[P_D(H)]_{\equiv \text{Pers}}.$$

Proof. The history already exists and is standing-bearing by theorem 11.1. Its retained consequence survives lawful continuation, its record change is not skin, and no same-act repair is permitted. These are the hypotheses of the imported persistence-realization theorem. The conclusion classifies the existing lineage; it does not generate it. $\square$

Part IV

Projection Fibres and Descent

13 Independently complete projection fibres

The four role fibres are formed before continuation is descended. In particular, the metric fibre is not defined by equality of untransported test pairings. Its profile is the lawful orbit of the complete natural test functional, so diffeomorphic representatives agree only after the tests and all source data are transported by the same lawful map.

13.1 Metric/source fibre

Definition 13.1 (Raw metric/source presentation and lawful groupoid). Fix the declared local space-time slab, dimension, signature, regularity class, boundary class, gauge class, compatible connection, and Cauchy scope. A raw metric/source presentation is

$$m = (g, \hat{x}, \text{Ward}, \text{Bdry}, \text{Gauge}, \mathcal{E}_g, d_g),$$

where g is the metric germ, $\hat{x}$ is an accepted Module A structured source object when source-bearing content is present, $\mathcal{E}_g$ is the registered hyperbolic system on the dynamical subtype, and d_g is its constraint-satisfying Cauchy-data class.

Let $\mathcal{G}_g \rightrightarrows \mathcal{M}_g$ be the lawful metric/source presentation groupoid. An arrow is a lawful diffeomorphism together with an accepted Module A source-improvement equivalence, preserving every fixed anchor and transporting the gauge, boundary, Ward, system, and Cauchy coordinates covariantly. This relation is fixed independently of the profile below.

Definition 13.2 (Natural metric/source test functional). For a raw presentation m , let Θ_m contain:

1. the complete fixed anchor, gauge, boundary, Ward, hyperbolic-system, and Cauchy-data ledgers;
2. the functional $h \mapsto \langle g, h \rangle$ on every compactly supported smooth symmetric tensor-density test h on the slab;
3. where source-bearing content is present, the accepted Module A attachment object and the functional $k \mapsto \langle \mathbf{T}_{\hat{x}}, k \rangle$ on every compact metric test tensor k ;
4. the metric-response role and declared source/exchange status.

The lawful groupoid acts on Θ_m by transporting both the presentation and its tests:

$$\langle \varphi^* g, h \rangle = \langle g, \varphi_* h \rangle, \quad \langle \varphi^* \mathbf{T}, k \rangle = \langle \mathbf{T}, \varphi_* k \rangle.$$

The metric profile is the orbit

$$\mathcal{P}_g(m) := [\Theta_m]_{\mathcal{G}_g}.$$

Theorem 13.3 (Metric probe completeness and lawful invariance). *For raw metric/source presentations m, n in the fixed target,*

$$\mathcal{P}_g(m) = \mathcal{P}_g(n) \iff m \text{ and } n \text{ are connected by a lawful arrow of } \mathcal{G}_g.$$

Moreover every standing-bearing metric/source use invariant under lawful redescription factors uniquely through the orbit carrier

$$Y_g := \mathcal{M}_g / \mathcal{G}_g.$$

Proof. A lawful arrow transports the metric, source, tests, anchors, Cauchy data, Ward and boundary ledgers together, so it preserves the orbit profile. Conversely, equality of orbit profiles means that for some lawful groupoid arrow the transported complete test functional of m equals that of n . Compact test pairings separate smooth metric germs and tensor distributions. The remaining anchor, gauge, boundary, Ward, hyperbolic-system and Cauchy coordinates are explicit components of the profile. Hence the transported raw presentations agree in every standing-bearing coordinate and are lawfully equivalent. The final factorization statement is the universal property of the orbit quotient. $\square$

Definition 13.4 (Metric quotient map and local gauge slice). Let

$$\Pi_g : U_D^g \twoheadrightarrow Y_g$$

be the restriction of the faithful upstream-to-metric projection to the local RPI chart. On a smooth Cauchy-eligible neighbourhood, a declared gauge-fixing certificate supplies a local section

$$\sigma_g : Y_g^{\text{Cau}} \longrightarrow \mathcal{M}_g, \quad \mathcal{P}_g^\sigma := \sigma_g \circ \Pi_g.$$

The section is used only to formulate a smooth Cauchy vector field; lawful metric standing remains the orbit class in Y_g .

13.2 Quantum fibre

Definition 13.5 (Quantum test profile). For a fixed complex Hilbert carrier and declared spatial envelope, the quantum profile $\mathcal{P}_H(x)$ contains:

1. the full ray transition function $\text{TP}([\psi_x], \cdot)$ against every ray;
2. the graph and domain class of the self-adjoint generator when a closed generator is claimed;
3. the spatial anchors, scalar potential/source ledger, homogeneous baseline, and closed/open/update status;
4. the full-ray process coordinate on the saturated subtype.

Basis and global phase are omitted as skin.

Theorem 13.6 (Quantum probe completeness). *Two quantum-role states have the same profile of definition 13.5 if and only if they are lawfully equivalent in the fixed projective-Hilbert target.*

Proof. If two rays have the same transition probability against every ray, testing against either ray gives transition value one, so the rays coincide. Equality of operator graphs gives equality of the closed operator and its domain. Equality of the spatial, potential, baseline, and closure ledgers fixes the remaining standing-bearing continuation data. Conversely, lawful unitary presentation change and global phase preserve all listed coordinates. Thus the profile separates exactly the fixed projective-Hilbert theorem object. $\square$

13.3 Record fibre

Definition 13.7 (Complete record profile). The record profile is the full retained, prospective, deliberative, and evidential reuse function together with fixation commutation, lineage, boundary trace, report, and failure status. Its carrier is the imported final quotient $\mathcal{A}_{\text{Ref}} / \simeq_{\text{MR}}$.

Theorem 13.8 (Record probe completeness). *Equality of the profile in definition 13.7 is necessary and sufficient for measurement-record standing equivalence in the declared use scope.*

Proof. The imported measurement-record quotient theorem states that every standing-bearing record presentation factors uniquely through equality of the complete reuse profile and fixation data. Distinct profiles cannot be identified without losing reference; equal profiles cannot be separated without a second same-scope classifier. Therefore the displayed profile is the final record fibre. $\square$

13.4 Compatibility fibre

Definition 13.9 (Complete compatibility profile). The compatibility profile contains the full fixed-scope witness ledger, the joint standing class, tensor constraints, local-realization family, boundary trace, no-transport statement, and separate coordinates for interaction, source transfer, signaling, branch selection, update, and repair.

Theorem 13.10 (Compatibility probe completeness). *Equality of the profile in definition 13.9 is necessary and sufficient for compatibility-role equivalence in the fixed scope.*

Proof. The imported compatibility theorem identifies the joint standing class with the complete witness content and excludes hidden signaling, mechanism, and repair typings on the same scope. The separate interaction and transfer coordinates ensure that genuine transport differences are not collapsed into compatibility. Hence equality of all listed coordinates is exactly lawful compatibility equivalence. $\square$

Definition 13.11 (Closed projection fibres). For $X \in \{g, H, \text{rec}, \text{comp}\}$, let

$$\Pi_X^0 : U_D^X \twoheadrightarrow Y_X$$

be the corresponding final closed-role quotient map. For notational continuity in Parts IV and V, Π_X denotes Π_X^0 . The exchange-extended maps $\tilde{\Pi}_X$ are different objects and are introduced only in Part VI.

Corollary 13.12 (No fourth role discriminator). *Within each fixed role target, a difference not detected by its independently final profile is skin; a detected difference is anchor or tensor content and cannot be removed by a same-scope selector.*

14 Smooth, strongly continuous, and exact projectability

The four role carriers require different continuation types. The regular metric Cauchy carrier is differentiable. The full-ray quantum carrier is treated by exact strongly continuous process descent, so an unbounded Stone generator is never used as a Banach vector field on the upstream chart. Record and compatibility carriers use exact or hybrid quotient preservation.

14.1 Smooth metric descent

Definition 14.1 (Smooth closed metric evolution). On a gauge-fixed smooth stratum of Y_g , the upstream seed is smoothly closed when there is a locally Lipschitz reduced metric/source vector field F_g such that

$$d\Pi_g(V_D) = F_g \circ \Pi_g$$

in a local representative chart of the lawful orbit quotient.

Theorem 14.2 (Smooth metric fibre preservation). *Smooth closed metric evolution implies that equal metric/source quotient classes remain equal for every common defined interval.*

Proof. Choose lawful representative charts for the common quotient class. Their projected curves solve the same locally Lipschitz reduced system with the same initial data. Uniqueness identifies the representative curves modulo the residual lawful groupoid, hence their quotient classes agree. $\square$

14.2 Exact quantum and quotient descent

Definition 14.3 (Exact closed role evolution). For $X \in \{H, \text{rec}, \text{comp}\}$, exact closure is the condition

$$\Pi_X(x) = \Pi_X(y) \implies \Pi_X(\alpha_{t,s}x) = \Pi_X(\alpha_{t,s}y)$$

for every common defined interval. For the quantum role the resulting process is additionally required to be strongly continuous on the full ray space. No derivative of the full-ray process is required at vectors outside a Stone-generator domain.

Theorem 14.4 (Exact fibre preservation). *Exact closure is precisely the condition required for the upstream flow to descend to the corresponding final quotient.*

Proof. The endpoint class depends only on the initial class. Thus the quotient assignment below is well defined. If exact closure failed, two representatives of one lawful class would produce different role classes, contradicting projectability. $\square$

Theorem 14.5 (Unique closed descended process). *If the smooth metric condition or an exact role condition applies, there is a unique local process*

$$\beta_{t,s}^X : Y_X \longrightarrow Y_X$$

satisfying

$$\Pi_X \circ \alpha_{t,s} = \beta_{t,s}^X \circ \Pi_X.$$

It obeys identity and cocycle laws on the declared domain. On the quantum branch it is strongly continuous whenever the exact closure certificate says so.

Proof. Define

$$\beta_{t,s}^X(\Pi_X x) = \Pi_X(\alpha_{t,s}x).$$

Fibre preservation gives well-definedness, surjectivity gives uniqueness, and the upstream flow laws descend. Strong continuity in the quantum role is inherited from the registered exact full-ray process, not from differentiability of the upstream seed in quantum directions. $\square$

Remark 14.6 (Role-specific closure modes). The metric profile uses a smooth local representative of the lawful orbit quotient. The quantum branch uses exact strongly continuous full-ray descent. Record and compatibility use exact or registered hybrid quotient closure. This type separation is essential for the full-ray inhabitant of Part VII.

15 Closed process categories and projection functors

The processes in this part are the *closed base processes*. They contain no declared exchange accumulator. Part VI constructs a distinct exchange-extended projection and actual enriched process.

Definition 15.1 (Descended fixed-clock closed process category). For each role X , let $\mathbf{Y}_X^{\tau,0}$ be the Moore-segment category of the closed process β^X , with morphisms

$$(y, T)_X : y \longrightarrow \beta_{T,0}^X(y),$$

quotiented only by lawful role-specific presentation equivalence preserving fixed clock duration.

Theorem 15.2 (Closed projection functor). *For every closed fibre, the quotient map induces a functor*

$$\Pi_X^0 : \mathbf{Cont}_D^\tau \longrightarrow \mathbf{Y}_X^{\tau,0}, \quad [(x, T)] \longmapsto (\Pi_X x, T)_X.$$

It preserves identities, composition, standing, failure, and clock orientation.

Proof. Fixed-clock lawful history equivalence preserves every final role class and duration. Identity and composition follow from the common duration and descended process cocycle. Standing and failure are retained upstream coordinates and cannot be changed by role quotienting. $\square$

Corollary 15.3 (Strict closed process square). *For every closed role X and every upstream morphism f ,*

$$\Pi_X \circ f = \beta^X(f) \circ \Pi_X$$

in the closed process category.

Proposition 15.4 (Clock-change compatibility of closed projection functors). *For an equivalent clock presentation, the duration-transport functors on the upstream and closed role categories make*

$$\begin{array}{ccc} \mathbf{Cont}_D^\tau & \xrightarrow{\Pi_X^0} & \mathbf{Y}_X^{\tau,0} \\ \mathfrak{T}_f \downarrow & & \downarrow \mathfrak{T}_f^X \\ \mathbf{Cont}_D^{\tau'} & \xrightarrow{(\Pi_X^0)'} & \mathbf{Y}_X^{\tau',0} \end{array}$$

commute. A clock-representative change is an equivalence between presentation categories, not an internal quotient of durations.

The equality in corollary 15.3 concerns the closed base projection. It does not imply that an exchange-extended actual endpoint must equal the counterfactual zero-exchange prediction. Those are separately typed objects in definitions 22.1 and 22.3.

Part V

Metric and Quantum Intertwining

16 Metric-source projection of an upstream history

Inherited governing use. The metric/source data in this part are the restriction of the already licensed faithful projection fixed by the dependency lock and imported theorem-use package, through the complete lawful profile of theorem 13.3 and definition 13.4. Metric field eligibility below is evaluated inside that same target. Its gravity role, fixed anchors, full same-scope first-response qualification, source-continuation obligations, homogeneous-vacuum quotient discipline, and coefficient/calibration envelope are inherited through the faithful representation; they are not licensed a second time by the

bookkeeping. The closed and declared-exchange alternatives retain the actual Module A attachment and Ward/exchange data. Neither source replacement nor cancellation of a current is supplied by a change of representation. Excluding an Einstein evolution from the definition means that no solved downstream history is used to form the upstream object; it does not discard the governing obligations already carried by the imported metric projection.

Definition 16.1 (Metric field eligibility). Let $U_D^g \subseteq U_D$ be the states whose final metric orbit contains a representative with:

1. the fixed four-dimensional Lorentzian anchors, local slab, regularity, gauge, boundary, and compatible connection;
2. a metric germ g and, where source-bearing content is present, an accepted Module A object $\hat{x} \in \hat{Q}_{\text{src}}$ with

$$q_g(\hat{x}) = [\mathbf{T}]_{\simeq_{\text{law}}};$$

3. closed Ward status $\nabla^\mu \mathbf{T}_{\mu\nu} = 0$ or an explicit external current J_ν ;
4. a same-scope first-response geometric candidate satisfying `SameScopeMetric1`;
5. sufficient regularity to evaluate the field equation distributionally or smoothly on the declared clock slab.

No Einstein evolution is included in this definition.

Definition 16.2 (Metric Cauchy eligibility). A metric field-eligible component is *Cauchy eligible* when it additionally carries:

1. a fixed smooth hyperbolic reduction $\mathcal{E}_g$ of the Einstein–matter system in a declared gauge;
2. constraint-satisfying projected initial data d_g ;
3. a registered smooth matter/source system whose Hilbert tensor is the accepted Module A source;
4. the local gauge-slice certificate of definition 13.4;
5. the infinitesimal seed-compatibility identity

$$\text{d}\mathcal{P}_g^\sigma(V_D) = \mathcal{F}_g \circ \mathcal{P}_g^\sigma$$

on the eligible RPI chart, where $\mathcal{F}_g$ is the independently defined reduced Einstein–matter Cauchy vector field;

6. the hypotheses of the local uniqueness theorem for $\mathcal{E}_g$.

These are dynamical certificates imposed on the already formed upstream seed. The gauge slice is a representative device inside one lawful orbit; it does not redefine metric standing or select a solved metric path.

Definition 16.3 (Projected metric history). For an upstream orbit contained in U_D^g , write

$$m_t := \Pi_g(\alpha_{t,0}x) \in Y_g.$$

On the Cauchy-eligible subtype, its declared gauge-slice representative is

$$(g_t, \mathbf{T}_t, \mathcal{B}_t, E_t) := \mathcal{P}_g^\sigma(\alpha_{t,0}x).$$

The metric-fibre theorem and smooth or exact closure give the descended closed process β^g on Y_g .

Proposition 16.4 (Module A ancestry is preserved). *If the initial metric orbit contains an attachment-certified Module A source and the metric fibre is closed under the upstream seed, every defined metric image remains in the same accepted source family or moves through an explicit exchange-extended coordinate introduced in Part VI.*

Proof. The final metric orbit retains the attachment graph, lawful source class, Ward current, support, regularity, boundary, Cauchy-system, and exchange status. Closed quotient descent prevents a representative of one orbit from silently changing those standing-bearing data. A changed source, Ward, boundary, or external-current coordinate is therefore detected and belongs to the enriched ledger rather than to lawful metric presentation equivalence. $\square$

Remark 16.5 (Distributional versus Cauchy branches). The pointwise field-equation theorem below is distributional where Module A supplies only a tensor distribution. Dynamical Cauchy intertwining is asserted only on the smooth Cauchy-eligible subtype of definition 16.2. No arbitrary distributional Einstein well-posedness theorem is claimed.

17 Einstein classification and genuine metric intertwining

Theorem 17.1 (Pointwise Einstein–Module-A normal form). *At every clock value of a metric field-eligible upstream history, the projected metric/source profile satisfies*

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = \kappa \mathbf{T}_{\mu\nu}$$

in the accepted regularity sense, with

$$\nabla^\mu \mathbf{T}_{\mu\nu} = 0$$

in closed scope and

$$\nabla^\mu \mathbf{T}_{\mu\nu} = J_\nu^{\text{declared}}$$

in declared-open scope.

Proof. Fix an arbitrary clock value. By definition 13.4 and theorem 13.3, the projected profile is the complete faithful representation of the same governing metric/source use fixed by the imported projection. The hypotheses of its Einstein normal-form theorem are inherited in this use: the gravity role, fixed metric anchors, full same-scope first-response and general-source qualification, actual source-continuation ledger, and homogeneous-vacuum quotient discipline. The accepted Module A attachment identifies the source in that governing relation with the same $\mathbf{T}$ represented in the profile; neither a different source nor a second conservation licence is introduced. Apply the imported Einstein normal-form theorem [6] in its original coefficient and calibration envelope. Its curvature-response exhaustion and divergence-compatibility result put the already received governing relation in the displayed form, while Module A supplies the actual closed or declared Ward identity. Faithful representation retains these obligations at every admitted clock value. This theorem classifies the projected field profile; it does not yet identify its parameterized path with a Cauchy development. $\square$

Definition 17.2 (Einstein Cauchy-flow category). For a fixed smooth hyperbolic system $\mathcal{E}_g$, let $\mathbf{Ein}_D$ have as objects constraint-satisfying gauge-fixed initial-data classes together with their accepted Module A source system, and as morphisms the clock-forward restrictions of the unique local Cauchy

development, modulo lawful metric redescription and accepted source improvement. Declared external currents are retained as directed exchange data.

Theorem 17.3 (Metric Cauchy intertwining). *On a Cauchy-eligible component, the descended metric process is the unique local Einstein–Module-A development of the projected initial data. Consequently the projection functor factors through $\mathbf{Ein}_D$ and, in closed scope,*

$$\Pi_g \circ f = \beta^g(f) \circ \Pi_g$$

with $\beta^g(f)$ the corresponding Cauchy morphism.

Proof. The seed-compatibility identity of Cauchy eligibility makes the projected profile curve an integral curve of the independently defined reduced Einstein–matter Cauchy vector field with initial data d_g . The local uniqueness theorem for the fixed system identifies the curves on their common slab [13, 14]. The strict process equation is already supplied by quotient descent, so the resulting morphism is the unique object of $\mathbf{Ein}_D$ with those data. In open scope the same statement holds only after the registered source exchange is included. $\square$

Corollary 17.4 (Vacuum identity branch). *For Minkowski initial data with the accepted Module A certified-zero source, zero cosmological term, and harmonic gauge, the descended metric process is the identity restriction of the unique local vacuum development.*

Proof. Minkowski spacetime is a smooth vacuum solution with the stated initial data. Local uniqueness identifies every projected solution with that development. This branch is used by the complete inhabitant in Part VI. $\square$

18 Projective quantum projection of the upstream flow

Definition 18.1 (Quantum-saturated upstream component). A quantum-eligible component $U_D^H \subseteq U_D$ is *saturated* when its complete quantum fibre is the full ray space $\mathbb{P}(\mathcal{H})$ of one declared complex Hilbert carrier and

$$\Pi_H : U_D^H \twoheadrightarrow \mathbb{P}(\mathcal{H})$$

is surjective.

Definition 18.2 (Transition-preserving full-ray closure). The seed is closed on the saturated quantum fibre when the exact quotient process β_t^H is defined on every ray in the common interval and

$$\mathrm{TP}(\beta_t^H[\psi], \beta_t^H[\phi]) = \mathrm{TP}([\psi], [\phi])$$

for all rays. The process is *Stone complete* when it has a strongly continuous all-real-parameter group extension agreeing with the local process.

Proposition 18.3 (Full-ray descended group). *On a saturated, transition-preserving, Stone-complete component, β_t^H is a strongly continuous one-parameter group of full-ray bijections preserving TP.*

Proof. Exact fibre closure gives the local quotient process. Saturation supplies the full ray carrier. Transition preservation is a registered equality on every ray pair, and Stone completeness supplies the all-real group extension, bijectivity, and strong continuity. The upstream history category remains clock forward even though the representative quantum group has inverses. $\square$

Definition 18.4 (Quantum spatial eligibility). The component is spatially eligible when the representative generator produced in the next section has a declared dense domain, is self-adjoint or essentially self-adjoint with certified closure, and satisfies the fixed scalar nonrelativistic SSQ_1 predicate: linearity, locality, Euclidean scalarity, uniform second-order principal symbol, declared scalar potential, and no hidden spin, gauge, relativistic, open, update, nonlocal, or boundary load.

19 Wigner–Stone–Schrödinger classification and quantum intertwining

Theorem 19.1 (Representative unitary flow). *On a saturated, transition-preserving, Stone-complete component, a continuous phase choice represents the ray process by a strongly continuous unitary group*

$$U_t = e^{-itH/\hbar}$$

with self-adjoint generator H .

Proof. Wigner’s theorem gives a unitary or antiunitary representative of each full-ray transition-preserving bijection. Continuity through the identity excludes the antiunitary branch. A continuous phase lift removes the scalar multiplier for the one-parameter translation group; any residual constant shift is the declared homogeneous energy baseline. Stone’s theorem gives the self-adjoint generator [15, 16, 17]. $\square$

Theorem 19.2 (Schrödinger spatial normal form). *On the spatially eligible subtype,*

$$H = -\frac{\hbar^2}{2m}\Delta + M_V$$

with its declared domain certificate, and

$$i\hbar\partial_t\psi_t = H\psi_t.$$

Proof. The representative generator exists by theorem 19.1. The imported Schrödinger first-response exhaustion theorem applies to the fixed scalar nonrelativistic envelope and leaves only the Laplacian and scalar multiplication slots. The admitted inertial/action anchors fix the coefficient. $\square$

Definition 19.3 (Schrödinger solution-segment category). Sch_D has as objects the declared projective-Hilbert systems with fixed generator/domain/source certificate and as morphisms the clock-forward restrictions of their Stone flow, modulo unitary presentation and global phase skin.

Theorem 19.4 (Quantum projection intertwining). *For every closed quantum-eligible upstream morphism,*

$$\Pi_H \circ f = \beta^H(f) \circ \Pi_H,$$

and $\beta^H(f)$ is the corresponding Schrödinger/Stone morphism. Measurement, bath, stochastic, or driven mismatch is represented only by a directed quantum exchange cell.

Proof. The strict equation follows from exact quotient descent. Theorems 19.1 and 19.2 independently classify the descended full-ray process. Open/update coordinates are separate from the closed quantum fibre and therefore cannot be hidden in unitary presentation equivalence. $\square$

Part VI

Record, Compatibility, and Directed Exchange

20 Record-fixation projection

Definition 20.1 (Upstream record quotient map). Let

$$\Pi_{\text{rec}} : U_D \longrightarrow \mathcal{A}_{\text{Ref}} / \simeq_{\text{MR}}$$

assign the final record-equivalence class determined by the complete reuse, fixation, trace, lineage, report, and failure profile of definition 13.7.

Definition 20.2 (Record-closed seed). The seed is record closed when it is exactly fibre preserving:

$$\Pi_{\text{rec}}(x) = \Pi_{\text{rec}}(y) \implies \Pi_{\text{rec}}(\alpha_{t,s}x) = \Pi_{\text{rec}}(\alpha_{t,s}y)$$

for every common defined interval. In declared update scope, the update event and its exchange incidence are separate coordinates of the upstream state.

Theorem 20.3 (Record process descent). *A record-closed seed induces a unique process*

$$\beta_{t,s}^{\text{rec}} : \mathcal{A}_{\text{Ref}} / \simeq_{\text{MR}} \rightarrow \mathcal{A}_{\text{Ref}} / \simeq_{\text{MR}}$$

and a functor

$$\mathbf{\Pi}_{\text{rec}} : \mathbf{Cont}_D^\tau \rightarrow \mathbf{Rec}_D$$

that preserves fixation, lineage, reuse, report, and failure status.

Proof. Record completeness is theorem 13.8. Exact fibre preservation gives well-defined quotient evolution by theorem 14.5. Because the listed standing-bearing coordinates define the final record quotient, the descended process preserves them or changes them only through a declared upstream update/exchange coordinate. $\square$

Corollary 20.4 (No record repair by later matching). *If a history uses a record before its fixation or lineage certificate exists, no later metric or quantum matching datum makes that earlier use the same admissible record act.*

21 Compatibility projection without transport laundering

Definition 21.1 (Upstream compatibility quotient map). Let

$$\Pi_{\text{comp}} : U_D \longrightarrow \mathcal{Q}_{\text{comp}}$$

assign the final compatibility class determined by the complete witness, joint-standing, boundary, local-realization, no-transport, interaction, source-transfer, update, and repair profile of definition 13.9.

Definition 21.2 (Compatibility-closed seed). The seed is compatibility closed when

$$\Pi_{\text{comp}}(x) = \Pi_{\text{comp}}(y) \implies \Pi_{\text{comp}}(\alpha_{t,s}x) = \Pi_{\text{comp}}(\alpha_{t,s}y)$$

for every common defined interval. Actual interaction, signal, source transfer, or update remains a separate upstream coordinate.

Theorem 21.3 (Compatibility process descent). *A compatibility-closed seed induces a unique process*

$$\beta_{t,s}^{\text{comp}} : \mathcal{Q}_{\text{comp}} \rightarrow \mathcal{Q}_{\text{comp}}$$

and a functor

$$\Pi_{\text{comp}} : \mathbf{Cont}_D^\tau \rightarrow \mathbf{Comp}_D.$$

In closed compatibility scope the joint standing class is continued as compatibility, not as metric communication or source motion.

Proof. Compatibility completeness is theorem 13.10. Exact fibre preservation gives quotient descent. Since interaction, transport, source transfer, and repair are separately retained coordinates, an unchanged compatibility class cannot acquire those roles through quotienting. $\square$

Proposition 21.4 (No compatibility laundering). *A change of metric source, record, branch, or local tensor content cannot be licensed merely by an unchanged compatibility class. It must occupy an interaction, exchange, update, or richer-scope coordinate.*

22 Closed prediction versus exchange-extended actual projection

The closed quotient processes of Part IV already satisfy strict descent. A nonzero exchange can therefore be represented only by a distinct exchange-extended projection and process. It is not a second reading of the same symbol β^X .

Definition 22.1 (Common incidence monoid and role restrictions). Let $(\mathcal{J}_D, \oplus, 0)$ be a conical commutative monoid of declared upstream exchange incidences. For each role $X \in \mathfrak{R}_4$, let

$$\rho_X : \mathcal{J}_D \longrightarrow \mathcal{J}_X$$

be a monoid homomorphism. The split exchange extension of the closed carrier is

$$\tilde{Y}_X := Y_X \times \mathcal{J}_X,$$

with inclusion $\iota_X(y) = (y, 0)$ and additive action

$$k \odot_X (y, j) = (y, k \oplus j).$$

The action category has an arrow labelled k from (y, j) to $(y, k \oplus j)$. Nonzero arrows need not be invertible.

Definition 22.2 (State ledger and common history incidence). An exchange-extended Module C certificate supplies an upstream ledger map

$$\ell_D : U_D \longrightarrow \mathcal{J}_D$$

and a history incidence

$$J_D : \text{Mor}(\mathbf{Cont}_D^\tau) \longrightarrow \mathcal{J}_D$$

such that for every $f : x \rightarrow y$,

$$\ell_D(y) = J_D(f) \oplus \ell_D(x),$$

with

$$J_D(\text{id}) = 0, \quad J_D(f_2 \circ f_1) = J_D(f_2) \oplus J_D(f_1).$$

The role ledger is $\ell_X = \rho_X \circ \ell_D$.

Definition 22.3 (Closed and actual projection maps). The closed role projection is the already constructed map

$$\Pi_X^0 := \Pi_X : U_D \rightarrow Y_X.$$

The exchange-extended actual projection is a different map,

$$\tilde{\Pi}_X : U_D \longrightarrow \tilde{Y}_X, \quad \tilde{\Pi}_X(x) = (\Pi_X^0 x, \ell_X(x)).$$

For a history f , define the lifted closed predictor and actual enriched process by

$$\hat{\beta}_X^0(f)(y, j) = (\beta^X(f)y, j),$$

$$\tilde{\beta}_X(f)(y, j) = (\beta^X(f)y, \rho_X(J_D(f)) \oplus j).$$

The first carries the incoming exchange ledger unchanged; the second advances it by the declared common incidence.

Proposition 22.4 (Two distinct intertwining statements). *For every closed-role fibre and every upstream history $f : x \rightarrow y$,*

$$\Pi_X^0(y) = \beta^X(f)(\Pi_X^0 x).$$

For every exchange-extended certificate,

$$\tilde{\Pi}_X(y) = \tilde{\beta}_X(f)(\tilde{\Pi}_X x).$$

The second equality does not redefine the first process: it follows from the first equality together with the independent ledger identity of definition 22.2.

Proof. The closed equality is the quotient-descent theorem. For the enriched equality, the base coordinate agrees by closed descent, while the ledger coordinate satisfies

$$\ell_X(y) = \rho_X(J_D(f)) \oplus \ell_X(x).$$

Thus the two coordinates agree separately. □

Definition 22.5 (Directed role exchange cell). For $f : x \rightarrow y$, the closed predicted endpoint in the enriched carrier is

$$z_X^0(f) := \widehat{\beta}_X^0(f)(\widetilde{\Pi}_X x),$$

and the actual endpoint is

$$z_X^1(f) := \widetilde{\Pi}_X y = \widetilde{\beta}_X(f)(\widetilde{\Pi}_X x).$$

The directed exchange cell is the action morphism

$$\mathcal{E}_X(f) : z_X^0(f) \longrightarrow z_X^1(f)$$

labelled by $\rho_X(J_D(f))$. In closed scope $\ell_D = 0$ and every cell is an identity.

Theorem 22.6 (Exchange identity and cocycle). *The exchange cells satisfy identity and source-target-compatible composition. For composable histories $f_1 : x \rightarrow y$ and $f_2 : y \rightarrow z$,*

$$\mathcal{E}_X(f_2 \circ f_1) = \mathcal{E}_X(f_2) \circ \widehat{\beta}_X^0(f_2)(\mathcal{E}_X(f_1)).$$

Proof. Identity follows from $J_D(\text{id}) = 0$. For composition, the first cell adds $\rho_X(J_D(f_1))$ to the ledger and is transported by the lifted closed predictor for f_2 without changing that ledger increment. The second cell adds $\rho_X(J_D(f_2))$. Their action-category composite adds

$$\rho_X(J_D(f_2)) \oplus \rho_X(J_D(f_1)) = \rho_X(J_D(f_2 \circ f_1)),$$

with the displayed order matching the source and target objects. □

Proposition 22.7 (Exchange is standing-bearing ledger content). *A nonidentity exchange changes a declared source, record, control, environment, boundary, or report ledger. It cannot be quotiented as coordinate, gauge, basis, phase, or other presentation skin.*

The split extension proves the frozen directed-exchange endpoint for declared ledger enrichment. A proposal in which the open interaction changes the closed base role itself must construct a different base process and is outside this split certificate rather than hidden in the ledger coordinate.

23 The exchange-certified four-role continuation cube

Let the ordered role set be

$$\mathfrak{R}_4 = \{g, H, \text{rec}, \text{comp}\}.$$

For one upstream history $f : x \rightarrow y$, each role has the closed-lift predicted endpoint $z_X^0(f)$ and the exchange-extended actual endpoint $z_X^1(f)$ of definition 22.5.

Definition 23.1 (Continuation four-cube). For every vertex

$$\varepsilon = (\varepsilon_g, \varepsilon_H, \varepsilon_{\text{rec}}, \varepsilon_{\text{comp}}) \in \{0, 1\}^4,$$

define

$$\mathcal{K}_f(\varepsilon) = \prod_{X \in \mathfrak{R}_4} z_X^{\varepsilon_X}(f)$$

in the product action category

$$\prod_{X \in \mathfrak{R}_4} (\tilde{Y}_X \rtimes \mathcal{J}_X).$$

An edge changing only coordinate X from 0 to 1 is the product of $\mathcal{E}_X(f)$ with identity morphisms in the other three coordinates.

Theorem 23.2 (Cubical face and cycle coherence). *The diagram*

$$\mathcal{K}_f : \{0, 1\}^4 \longrightarrow \prod_X (\tilde{Y}_X \rtimes \mathcal{J}_X)$$

is a well-defined four-cube. Every two-dimensional face commutes, and every two monotone paths with the same endpoints have the same composite. In closed scope all edges are identities and the cube commutes strictly.

Proof. Exchange edges in different role coordinates act in different factors of a product category. Hence the interchange law gives

$$(\mathcal{E}_X \times \text{id}) \circ (\text{id} \times \mathcal{E}_Y) = (\text{id} \times \mathcal{E}_Y) \circ (\mathcal{E}_X \times \text{id}).$$

Every square face therefore commutes. Any two monotone paths differ by a finite sequence of adjacent swaps of independent coordinate edges, so their composites agree. When $J_D(f) = 0$, every role restriction is zero and all edges are identities. $\square$

Theorem 23.3 (Functorial cube assignment). *For composable upstream histories, the cube assignment respects composition. The long-history cube is obtained by whiskered coordinatewise pasting of the two short-history cubes, and every role edge remains the restriction of the same common incidence assignment.*

Proof. The closed role processes compose by their descended cocycles. The enriched actual processes compose by additivity of J_D . The exchange cocycle of theorem 22.6 composes each coordinate edge. Whiskered coordinatewise pasting in the product action category therefore gives the cube of the composite history. $\square$

Corollary 23.4 (Closed continuation cube). *If $J_D(f) = 0$, then for all four roles*

$$\Pi_X^0 \circ f = \beta^X(f) \circ \Pi_X^0,$$

and every face and cycle of the four-cube commutes strictly.

Corollary 23.5 (No deferred cube repair). *A missing attachment, failed record fixation, hidden source transfer, or unregistered exchange on a prefix cannot be repaired by a later cube while preserving the same upstream history.*

Part VII

Inhabited Continuation Packets and Bounded Audit Artifact

24 A complete closed nonidentity inhabitant

The complete inhabitant uses a globally smooth upstream seed. The unbounded Stone generator appears only in the clock-conditioned downstream quantum projection; it is not treated as a Banach vector field on the full projective carrier.

Definition 24.1 (Fixed local RPI laboratory packet). Fix a bounded laboratory world tube

$$W_\star = (-\varepsilon, \varepsilon) \times [-L, L]$$

with a local Minkowski comparison presentation. Let $\mathcal{Q}_\star$ be the full abstract projective predictive-standing carrier of a stable scalar role. Within the regular local RPI chart, $\mathcal{Q}_\star$ is equipped with the standard smooth projective-Hilbert-manifold structure inherited from the chosen faithful projective-Hilbert representative chart. This smooth structure is used only to certify local chart regularity and carries no upstream authority over the kernel, the realized physical interior, or the continuation seed. Let $\mathcal{R}_\star \cong \mathbb{R}$ be a retained record coordinate. The local RPI chart is

$$U_\star = J \times \mathcal{Q}_\star \times \mathcal{R}_\star, \quad x = (\theta, q, r),$$

with fixed role tuple

$$\mathfrak{I}_\star = (\text{NonVacRec}_\star, B_\star, \text{LCD}_\star, \text{Cmp}_\star, \text{BoundTrans}_\star, \Phi_\star, \Pi_\star, G_\star, M_\star, \Omega_\star, T_\star, \text{Cau}_\star).$$

Here $B_\star = [-L, L]$, $\text{LCD}_\star$ is its local subinterval family, $\Phi_\star$ is the scalar carrier, $\Pi_\star$ its localized persistence class, $G_\star$ the phase-presentation quotient, $M_\star = m > 0$ the inertial response anchor, $\Omega_\star$ the scalar/vacuum baseline, and $T_\star, \text{Cau}_\star$ the retained clock order and witnessed dependence relation. The standing locus is the whole chart while $\theta \in J$; exit from J is the registered failure boundary.

Proposition 24.2 (RPI-role and clock adequacy). *The packet of definition 24.1 is a nondegenerate local occupant of every RPI role required by the Module C target. The coordinate*

$$\tau_\star(\theta, q, r) = \theta$$

together with the retained record r , orientation $d\theta > 0$, interval J , and boundary failure is a certified local physical-clock representative for the declared continuation use.

Proof. The bounded cell and its subintervals supply localization and local continuation domains. The projective predictive carrier supplies quotient-stable comparison and a non-null standing-bearing

state role; the scalar carrier, persistence class, phase quotient, inertial response, and baseline occupy the field, particle, gauge, mass, and vacuum roles. The record coordinate and clock order retain irreversible consequence, and later record values depend on earlier packet states through the admitted flow. All entries share one carrier, standing predicate, lawful quotient, and continuation envelope. Since $d\tau_\star(\partial_\theta) = 1$, the clock is forward on every nontrivial segment, while exit from J is recorded as failure rather than silently continued. $\square$

Definition 24.3 (Globally smooth upstream seed). Define

$$V_\star = \partial_\theta + \partial_r$$

on $U_\star$. Its local flow is

$$\alpha_t^\star(\theta, q, r) = (\theta + t, q, r + t)$$

whenever the clock interval remains inside J .

Proposition 24.4 (Seed formation, regularity, and nonidentity). *The seed of definition 24.3 is C^∞ , locally Lipschitz, clock normalized, standing tangent, and formed without any metric, Hilbert generator, or solved downstream path. Every positive-duration segment is nonidentity.*

Proof. The vector field is constant in the regular product chart and has zero component in the full projective coordinate. Hence it is globally smooth on the local slab and its flow is explicit. The clock and retained record both advance by t , so a positive-duration morphism cannot be an identity even when every downstream dynamical profile is static. $\square$

Definition 24.5 (Clock-conditioned full-ray quantum projection). Let $(R_t^\star)_{t \in \mathbb{R}}$ be the abstract full-ray transition-preserving group with spectral coordinates

$$R_t^\star[c_0, c_1, \dots] = [e^{-iE_0 t/\hbar} c_0, e^{-iE_1 t/\hbar} c_1, \dots].$$

A faithful projective-Hilbert representative chart identifies it downstream with the Stone group of the self-adjoint harmonic-oscillator Hamiltonian

$$H_\star = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2$$

on its standard domain. Define the four closed projections only after the upstream seed has been fixed:

$$\begin{aligned} \Pi_g^0(\theta, q, r) &= m_0, & m_0 &= [(\eta, \mathbf{0}, \text{vacuum Cauchy data})]_g, \\ \Pi_H^0(\theta, q, r) &= R_\theta^\star q, \\ \Pi_{\text{rec}}^0(\theta, q, r) &= [r]_{\text{MR}}, \\ \Pi_{\text{comp}}^0(\theta, q, r) &= c_0, \end{aligned}$$

where c_0 is the fixed factorized compatibility class with explicit no-transport and no-source-transfer coordinates.

Proposition 24.6 (Strict projection processes). *For the upstream flow of definition 24.3, the descended*

role processes are

$$\begin{aligned}\beta_g^t(m_0) &= m_0, \\ \beta_H^t(q') &= R_t^* q', \\ \beta_{\text{rec}}^t([r]_{\text{MR}}) &= [r + t]_{\text{MR}}, \\ \beta_{\text{comp}}^t(c_0) &= c_0.\end{aligned}$$

They satisfy strict intertwining with the four closed projections.

Proof. The metric and compatibility equalities are immediate. For the quantum branch,

$$\Pi_H^0(\alpha_t^*(\theta, q, r)) = R_{\theta+t}^* q = R_t^*(R_\theta^* q) = \beta_H^t(\Pi_H^0(\theta, q, r)).$$

The record equality is the translation identity. Since q ranges over the full projective carrier and R_θ^* is bijective, Π_H^0 is saturated. The group is strongly continuous and transition preserving; Wigner and Stone give its unitary representative and generator, and the accepted spatial normal-form theorem yields the displayed Schrödinger Hamiltonian. The metric branch is the unique local Minkowski vacuum development with the accepted Module A certified-zero source. $\square$

Theorem 24.7 (Complete closed-packet verification). *The packet of definitions 24.1, 24.3 and 24.5 satisfies C1–C12, contains nonidentity upstream morphisms, and yields a strict commuting four-cube.*

Proof. C1–C5 follow from propositions 24.2 and 24.4. The four fibres are the independently final carriers of Part IV. The metric branch is constant and Cauchy eligible; the quantum branch is exact on the full ray carrier and Stone complete; the record and compatibility branches preserve their final quotient fibres. Thus C6–C10 hold. Set the common ledger identically to zero. Then every enriched projection equals the zero-ledger inclusion of its closed projection, every exchange cell is an identity, and theorem 23.2 gives C11. Positive-duration histories change both the clock and record, so C12 holds. The bounded certificate records two composable segments and all sixteen strict vertices. $\square$

The unbounded oscillator generator is used only in the downstream Stone classification of the full-ray process. The upstream seed remains the globally smooth field $\partial_\theta + \partial_r$. The packet is a local strict continuation certificate, not a nonzero interacting quantum-gravity model.

25 A complete declared-open packet with distinct prediction and actual projection

The declared-open example uses the same smooth upstream seed pattern as the closed packet and adds one explicit upstream ledger coordinate. The closed metric prediction remains the vacuum profile; the actual exchange-extended metric projection additionally retains the accumulated omission ledger. These are different maps and different processes.

Definition 25.1 (Common-incidence open packet). Let

$$U_\star^{\text{open}} = J \times \mathcal{Q}_\star \times \mathcal{R}_\star \times \mathbb{R}_{\geq 0}, \quad x = (\theta, q, r, e),$$

with upstream seed and flow

$$V_{\star}^{\text{open}} = \partial_{\theta} + \partial_r + \partial_e,$$

$$\alpha_t^{\text{open}}(\theta, q, r, e) = (\theta + t, q, r + t, e + t).$$

The common incidence monoid is

$$\mathcal{J}_D = (\mathbb{R}_{\geq 0}, +, 0),$$

with role restrictions

$$\rho_g(j) = j, \quad \rho_H(j) = \rho_{\text{rec}}(j) = \rho_{\text{comp}}(j) = 0.$$

The upstream ledger is $\ell_D(\theta, q, r, e) = e$, so a duration- t history carries $J_D(f_t) = t$.

Definition 25.2 (Closed and enriched projections of the open packet). The four closed projections are exactly those of definition 24.5 and do not contain e :

$$\Pi_g^0 = m_0, \quad \Pi_H^0 = R_{\theta}^{\star} q, \quad \Pi_{\text{rec}}^0 = [r]_{\text{MR}}, \quad \Pi_{\text{comp}}^0 = c_0.$$

The exchange-extended actual projections are

$$\begin{aligned} \tilde{\Pi}_g(\theta, q, r, e) &= (m_0, e), \\ \tilde{\Pi}_H(\theta, q, r, e) &= (R_{\theta}^{\star} q, 0), \\ \tilde{\Pi}_{\text{rec}}(\theta, q, r, e) &= ([r]_{\text{MR}}, 0), \\ \tilde{\Pi}_{\text{comp}}(\theta, q, r, e) &= (c_0, 0). \end{aligned}$$

The metric ledger coordinate is the declared fixed-background omission. It is not part of the lawful metric orbit m_0 and is not used to redefine the closed metric process.

Proposition 25.3 (Open-packet predicted and actual processes). *For a duration- t history, the lifted closed metric predictor and actual enriched metric process are*

$$\begin{aligned} \hat{\beta}_g^0(t)(m_0, e) &= (m_0, e), \\ \tilde{\beta}_g(t)(m_0, e) &= (m_0, e + t). \end{aligned}$$

The other three enriched processes have zero exchange restriction and are the zero-ledger lifts of their strict closed processes. Each actual enriched projection intertwines with the corresponding actual enriched process.

Proof. The base metric orbit is static in both processes. The closed predictor leaves the incoming omission ledger unchanged, whereas the actual process advances it by the registered common incidence. The equality

$$\tilde{\Pi}_g(\alpha_t^{\text{open}} x) = (m_0, e + t) = \tilde{\beta}_g(t)(\tilde{\Pi}_g x)$$

is immediate. The quantum, record, and compatibility equalities are the strict identities from proposition 24.6 with a zero ledger coordinate appended. $\square$

Proposition 25.4 (Open-packet exchange cocycle and cube). *For composable duration-one histories f_{01} and f_{12} and their duration-two composite f_{02} ,*

$$J_D(f_{02}) = J_D(f_{12}) + J_D(f_{01}) = 2.$$

The metric exchange cell adds the corresponding omission ledger, the other three role cells are identities, and the sixteen-vertex cubes compose functorially.

Proof. The metric exchange cell is the action morphism

$$(m_0, e) \longrightarrow (m_0, e + t)$$

labelled by t . Additivity of the ledger gives the correctly typed cocycle of theorem 22.6. The other role restrictions vanish. Therefore all four cells arise from one common incidence assignment, and theorems 23.2 and 23.3 give face, cycle, and composition coherence. $\square$

Theorem 25.5 (Complete declared-open inhabitation). *The packet of definitions 25.1 and 25.2 satisfies every applicable coordinate C1–C12 and inhabits the exchange-certified four-cube endpoint.*

Proof. The RPI and clock roles are the closed packet roles with one additional declared ledger coordinate. The seed is constant, globally smooth on the local slab, clock normalized, standing tangent, and nonzero. The four closed fibres and their strict processes are those already verified in theorem 24.7. The actual enriched projections and processes are separately constructed in definition 25.2 and proposition 25.3. Common incidence, role restriction, identity, and composition are proved in proposition 25.4. Positive-duration histories change clock, record, and omission ledger, so the packet is nonidentity. The finite certificate contains two composable open histories and their sixteen predicted/actual vertices. $\square$

The omission ledger records that the oscillator is treated on a fixed background in this comparison packet. It does not claim that the background metric is sourced by the oscillator, and it does not turn the fixed-background construction into a closed Einstein–quantum coupling.

26 Bounded certificate checker

The project contains a deterministic checker for the finite registry layer. It does not verify Banach-flow existence, RPI adequacy, probe completeness, Einstein or Schrödinger equations, self-adjointness, or the analytic exchange action.

Definition 26.1 (Finite continuation certificate). A machine-readable certificate contains:

1. accepted Module A–B artifact hashes;
2. a fixed clock-representative identifier;
3. upstream states with standing, clock, record, metric, quantum, compatibility, failure, and tensor-signature hashes;
4. morphisms with source, target, duration, legal/coherent status, and a complete finite path-sample registry for the encoded claim;
5. predicted and actual endpoints for all four roles;
6. all sixteen cubical vertex hashes;
7. a common-incidence registry, one incidence amount per open morphism, and role restriction weights;

8. role exchange identifiers, endpoints, amounts, and incidence membership;
9. composition triples and expected exchange cocycles;
10. at least one nonidentity upstream morphism.

Proposition 26.2 (Software correspondence). *If the checker returns **PASS**, then the encoded finite registry has exact references, standing at every registered path sample, clock-forward and duration-additive composition, correctly generated sixteen-vertex cubes, declared exchange incidence drawn from the registry, correct role restrictions, exact exchange cocycles on every registered composition, and at least one nonidentity morphism.*

Proof. Each conclusion is a direct validation branch of `software/certificate.py`. The claim is intentionally restricted to registered finite data. Analytic standing between registered samples and the physical adequacy of the mathematical objects remain manuscript theorems. $\square$

The tests include valid strict and open certificates and failures for clock reversal, failed path samples, missing incidence membership, incorrect role restriction, incomplete cube vertices, endpoint mismatch, nonadditive duration, broken exchange cocycle, and identity-only certificates.

Part VIII

Endpoint, Capstone Handoff, and Adversarial Closure

27 Main standalone theorems

Theorem 27.1 (Kernel-native upstream continuation construction). *Every complete Module C certificate constructs an independently formed nonempty fixed-clock upstream continuation category $\mathbf{Cont}_D$ before any downstream process is invoked. It is clock forward, coherent under prefix and composition, invariant under fixed-clock lawful redescription, fail closed against same-lineage repair, and presentation equivalent to the category obtained from any clock-equivalent representative through explicit duration transport.*

Proof. Combine theorems 7.1, 8.1, 8.4, 10.5 and 11.1. Nonidentity and a globally smooth full-ray-compatible seed are supplied by theorem 24.7. $\square$

Theorem 27.2 (Four complete closed fibres and role-process descent). *The metric, quantum, record, and compatibility fibres of Part IV are complete in their declared scopes. Smooth metric closure or exact role closure therefore induces unique closed role processes and projection functors. On the eligible subtypes:*

1. the metric process is the unique local Einstein–Module-A Cauchy development;
2. the quantum process is an exact strongly continuous full-ray Wigner–Stone–Schrödinger history;

3. the record process preserves the final fixation/reuse/lineage quotient;
4. the compatibility process preserves the final joint-standing class without transport laundering.

Proof. Metric finality is the lawful groupoid-orbit theorem theorem 13.3; the other finalities are theorems 13.6, 13.8 and 13.10. Descent is theorems 14.5 and 15.2. Role classification is theorems 17.3, 19.4, 20.3 and 21.3. $\square$

Theorem 27.3 (Distinct closed and exchange-extended processes). *For every declared-open certificate, the closed prediction $(\Pi_X^0, \hat{\beta}_X^0)$ and actual exchange-extended process $(\tilde{\Pi}_X, \tilde{\beta}_X)$ are independently defined and satisfy their own intertwining equations. The directed exchange cell adds precisely the role restriction of the common incidence and obeys identity and source-target-correct composition.*

Proof. This is definitions 22.3 and 22.5, proposition 22.4, and theorem 22.6. The closed quotient descent controls the base coordinate; the upstream ledger identity controls the enriched coordinate. $\square$

Theorem 27.4 (Strict and exchange-certified cubical closure). *For every full upstream morphism with complete projection certificates, the sixteen predicted/actual role vertices form the four-cube of definition 23.1. In closed scope every edge is an identity. In declared-open scope every edge is the corresponding restriction of one common incidence. Every face, monotone path, and registered composition commutes.*

Proof. Apply theorems 23.2, 23.3 and 27.3. $\square$

Corollary 27.5 (Standalone Module C endpoint). *The unchanged frozen standalone endpoint is inhabited by:*

1. the complete closed clock-translation packet of theorem 24.7; and
2. the complete declared-open common-incidence packet of theorem 25.5.

Both are formed upstream of their metric, Hilbert, record, and compatibility classifications. The unbounded quantum generator appears only downstream in a full-ray relational projection, and the nonzero exchange cube compares distinct closed and actual projections rather than two readings of one descended process.

28 Capstone handoff

The accepted output sought from Module C is the following theorem-use package:

1. one independently constructed fixed-clock RPI continuation category;
2. explicit equivalences between clock-equivalent presentations, without collapsing their duration coordinates;
3. four independently final role fibres and unique descended closed processes;
4. genuine local Einstein Cauchy intertwining on the smooth gauge-slice metric subtype;
5. Wigner–Stone–Schrödinger intertwining on the saturated full-ray quantum subtype;
6. final record and compatibility quotient descent;

7. a strict distinction between the closed projection/process pair and the exchange-extended actual projection/process pair;
8. one common directed exchange-incidence object and a coherent sixteen-vertex four-cube;
9. one complete strict nonidentity inhabitant and one complete declared-open exchange inhabitant.

Module D may use these objects only after a positive standalone Module C review. It may not replace them by generic pullback, colimit, role-label, or downstream-synchronization arguments.

29 Adversarial map

Objection	Required response or failure locus
The seed is a solved downstream path in disguise.	Refuted by the RPI chart, fixed clock, and smooth translation seed being formed before every role projection. Failure would occur at definitions 6.1 and 24.3.
The vector field is generated by the kernel.	Not claimed. It is explicit tensor content in an already admitted interior.
Clock reparametrization is ill typed.	One representative is fixed per Moore category. Another representative produces a separate category related by the duration-transport equivalence of theorem 10.5.
The metric profile is not diffeomorphism invariant.	Repaired: the profile is the groupoid orbit of the entire natural metric/source test functional, with tests and all ledgers transported together. Equality of orbit profiles is equivalent to lawful metric/source equivalence by theorem 13.3.
Projection completeness is assumed.	Each role has a separate orbit/separation/final-quotient theorem in Part IV. A criticism must break one of theorems 13.3, 13.6, 13.8 and 13.10.
Discrete records are differentiated.	They are not. Record and compatibility use exact or registered hybrid fibre preservation.
The metric branch is only stagewise Einstein.	Pointwise normal form and Cauchy intertwining are separated. The process theorem uses a fixed hyperbolic system, a local gauge section, and local uniqueness.
The full-ray quantum packet has no smooth upstream seed because H is unbounded.	Repaired: the upstream seed is the constant field $\partial_\theta + \partial_r$ and has no quantum component. The unbounded Stone group appears only in the clock-conditioned projection $\Pi_H(\theta, q, r) = R_\theta q$.
A finite ray path is insufficient for Wigner–Stone.	Correct. The theorem requires a saturated full-ray, transition-preserving, strongly continuous all-real group.
Record or compatibility is used as hidden transport.	Their final quotients retain update, interaction, source-transfer, and repair as separate coordinates.

The exchange contradicts quotient descent.	Repaired: $(\Pi_X^0, \hat{\beta}_X^0)$ is the closed prediction, while $(\tilde{\Pi}_X, \tilde{\beta}_X)$ is the actual exchange-extended projection. They are different maps and processes; the exchange cell relates their independently typed enriched endpoints.
Exchange is merely a label.	The common incidence is a conical monoid acting in explicit enriched role categories. State ledgers, role restrictions, actual processes, identity, and source-target-correct cocycles are constructed in Part VI.
Four squares are not a cube.	The cube has sixteen vertices indexed by $\{0, 1\}^4$, coordinate edges, commuting two-faces, monotone path-independence, and functorial composition.
The closed inhabitant is incomplete.	The packet supplies the complete local RPI tuple, clock certificate, smooth nonidentity seed, final four fibres, vacuum metric Cauchy branch, full-ray clock-conditioned quantum branch, record/compatibility quotients, and zero incidence.
The open inhabitant reuses the same inconsistent process.	Repaired: it adds an upstream ledger coordinate and uses distinct closed and enriched projections. Two composable histories verify the nonzero metric exchange cocycle.
The software proves the analytic theorem.	It does not. Its proposition is restricted to registered state/path samples, separate base and enriched endpoints, incidence restrictions, cube vertices, and additive cocycles.
The module proves interacting quantum gravity.	It does not. The open metric ledger explicitly records a fixed-background omission rather than claiming closed backreaction.

Theorem 29.1 (Same-scope failure exhaustion). *A proposed same-target Module C rival that fails the construction falls into at least one of:*

1. *downstream-pullback formation;*
2. *clock/seed conflation or cross-category duration collapse;*
3. *missing local-flow existence or nonvacuity;*
4. *standing/no-repair failure;*
5. *failure of a lawful-orbit or final-quotient fibre theorem;*
6. *failure of smooth metric or exact quantum/record/compatibility projectability;*
7. *missing metric Cauchy, full-ray quantum, record, or compatibility eligibility;*
8. *conflation of closed prediction with exchange-extended actual projection;*
9. *hidden, unregistered, or noncomposable exchange incidence;*
10. *noncubical or noncoherent role matching;*
11. *incomplete inhabitance, presentation skin, scope change, or standing failure.*

Proof. Take the first certificate coordinate at which the rival differs. Every endpoint coordinate now has an independently typed construction and local completeness theorem. A difference changing none

of them changes no standing-bearing use and is skin; an undeclared difference changing one of them is the corresponding listed failure or a scope change. $\square$

30 Conclusion

Module C constructs the missing continuation object without reversing the kernel dependency order. One certified local clock representative and one explicit RPI seed produce a nonempty fixed-clock Moore-history category before metric, quantum, record, or compatibility processes are evaluated. Clock-equivalent representatives yield duration-transport equivalent categories rather than ill-typed automorphisms of one category.

The metric fibre is the lawful groupoid orbit of the complete natural metric/source functional, not equality of untransported representative test values. The quantum fibre is fixed by full transition, process, generator-domain, and spatial/source data; record and compatibility are their final imported quotients. The regular metric branch descends through a local gauge section and is Cauchy-intertwined with the Einstein–Module-A system. The quantum branch uses exact strongly continuous full-ray descent; its unbounded Stone generator is downstream and never acts as the upstream Banach vector field.

Open continuation is also type-correct. The closed prediction and actual exchange-extended projection are separate maps and processes. A common upstream ledger advances only enriched role coordinates, and the directed action cells satisfy identity and source-target-correct cocycle laws. Their product forms a genuine sixteen-vertex four-cube with commuting faces, paths, and compositions.

The closed packet uses the globally smooth seed $\partial_\theta + \partial_r$ and the relational quantum projection $\Pi_H(\theta, q, r) = R_\theta q$. It supplies a strict nonidentity cube. The open packet adds one smooth ledger coordinate and supplies two composable nonzero metric-exchange segments without redefining the closed vacuum process.

The final theorem-use output is therefore

$$\begin{aligned} \text{one upstream RPI continuation} &\implies \text{four complete closed role processes} \\ &\implies \text{one strict or exchange-certified actual four-cube.} \end{aligned}$$

It is not dynamics generated by the kernel, not a global clock theorem, not a nonzero interacting quantum-gravity law, and not the capstone unique orbit. Those boundaries remain fixed for later modules.

A Formal predicates

A.1 Fixed-clock upstream seed

Write

$$\text{Seed}_D^\tau(U, \tau, V, \Sigma_I)$$

when:

1. U is a regular local chart of the lawful RPI quotient with a complete RPI role packet;
2. τ is one certified oriented local clock representative for the declared use;
3. V is C^1 , locally Lipschitz, clock normalized, standing tangent, and nonzero somewhere;

4. failure and no-repair data are fail closed;
5. any common exchange coordinate is explicit upstream tensor data;
6. no downstream metric, Hilbert, record, compatibility, or solved path is used to define V .

A.2 Clock-presentation transport

For $\tau' = f \circ \tau$ and $V' = (f' \circ \tau)^{-1}V$, set

$$\text{TimeChange}_f(x, T) = f(\tau(x) + T) - f(\tau(x)).$$

Clock-presentation invariance means an equivalence between separately constructed Moore categories using this duration transport. It does not identify unequal durations inside one category.

A.3 Final role fibres

For the metric role, let $\mathcal{G}_g \rightrightarrows \mathfrak{M}_g$ be the lawful metric/source groupoid and define

$$Y_g = \mathfrak{M}_g / \mathcal{G}_g.$$

Metric completeness is equality of lawful orbit profiles of the complete natural metric/source functional. For the other roles, Y_H , Y_{rec} , and Y_{comp} are their stated final quotients. Write

$$\text{CompleteFib}_X(\Pi_X)$$

when equality in Y_X is necessary and sufficient for lawful role equivalence.

A.4 Projectability

The regular metric branch is smoothly closed when, in a declared local gauge section,

$$\text{d}\Pi_g(V) = F_g \circ \Pi_g$$

for a locally Lipschitz reduced metric/source field F_g .

For $X \in \{H, \text{rec}, \text{comp}\}$, exact closure is

$$\Pi_X(x) = \Pi_X(y) \implies \Pi_X(\alpha_t x) = \Pi_X(\alpha_t y).$$

For $X = H$, the descended process must additionally be a strongly continuous full-ray group on the quantum-eligible subtype. No unbounded generator is required to define the upstream seed.

A.5 Metric and quantum eligibility

$\text{MetricCauchyElig}(x)$ means: fixed Lorentzian anchors, accepted Module A source, lawful metric orbit class, smooth local gauge section, registered hyperbolic Einstein–matter reduction, constraint-satisfying data, infinitesimal seed compatibility, and local uniqueness.

$\text{QuantumElig}(x)$ means: full-ray saturation of Π_H , exact transition preservation, a strongly continuous all-real group, and a downstream representative generator/domain satisfying SSQ_1 .

A.6 Closed and exchange-extended processes

Let

$$\Pi_X^0 : U_D \rightarrow Y_X, \quad \tilde{\Pi}_X : U_D \rightarrow Y_X \times \mathcal{J}_X$$

be distinct maps. The closed predictor and actual enriched process are

$$\hat{\beta}_X^0(f)(y, j) = (\beta_X(f)y, j),$$

$$\tilde{\beta}_X(f)(y, j) = (\beta_X(f)y, j \oplus \rho_X J_D(f)).$$

The directed exchange cell is the action morphism from the first endpoint to the second.

A.7 Exchange and cube

The common incidence satisfies

$$J_D(\text{id}) = 0, \quad J_D(f_2 \circ f_1) = J_D(f_2) \oplus J_D(f_1).$$

For initial ledger j ,

$$\mathcal{E}_X(f_2 \circ f_1) = \mathcal{E}_X(f_2) \circ \hat{\beta}_X^0(f_2)(\mathcal{E}_X(f_1)).$$

$\text{CubeClosed}(f, j)$ means the functor from $\{0, 1\}^4$ to the product action category has all sixteen vertices, all coordinate edges, commuting square faces, path-independence, and functorial composition.

A.8 Module C endpoint

$$\begin{aligned} \text{ModuleCClosed}(\mathcal{C}_D^C) &:= \text{Seed}_D^\tau \wedge \text{FlowExists} \wedge \text{FixedClockHistoryCategory} \wedge \text{Nonidentity} \\ &\wedge \bigwedge_X \text{CompleteFib}_X \wedge \text{MetricCauchyIntertwine} \wedge \text{QuantumIntertwine} \\ &\wedge \text{RecordIntertwine} \wedge \text{CompatIntertwine} \wedge \text{DistinctClosedActualExchange} \\ &\wedge \text{ExchangeCube} \wedge \text{ClosedInhabitant} \wedge \text{OpenInhabitant}. \end{aligned}$$

B Flow, time change, and projectability details

B.1 Local-flow construction

Let $\varphi : U_0 \rightarrow E$ be a Banach or Hilbert-manifold chart and write

$$\tilde{V} = d\varphi \circ V \circ \varphi^{-1}.$$

Local Lipschitz continuity gives a contraction map

$$(\mathcal{T}\gamma)(t) = \varphi(x) + \int_0^t \tilde{V}(\gamma(s)) \, ds$$

on a sufficiently small closed ball. Its fixed point is the unique integral curve. The flow law follows from uniqueness after time translation. In the explicit inhabitants the seed is globally smooth on the declared local product manifold and the flow is written in closed form.

B.2 Standing submanifold invariance

If $\mathcal{S}_D$ is embedded and $V_x \in T_x \mathcal{S}_D$, the restricted field has a local flow on $\mathcal{S}_D$. Inclusion gives an ambient integral curve with the same initial value, so uniqueness identifies the two.

B.3 Fixed-clock Moore composition

A τ -Moore segment retains its duration. For

$$x \xrightarrow{T} y \xrightarrow{S} z,$$

the composite is $x \xrightarrow{T+S} z$. Strict associativity follows from addition.

B.4 Clock-presentation functor

For $\tau' = f \circ \tau$, the same path has duration

$$\rho_f(x, T) = f(\tau(x) + T) - f(\tau(x)).$$

The identity

$$\rho_f(x, T + S) = \rho_f(x, T) + \rho_f(\alpha_T x, S)$$

proves functoriality. Clock changes therefore relate separate categories by equivalence; they do not generate an internal duration quotient.

B.5 Smooth metric descent

In a declared local gauge section of the lawful metric orbit carrier, if

$$d\Pi_g(V) = F_g \circ \Pi_g$$

with F_g locally Lipschitz, equal initial metric profiles solve the same reduced system and remain equal. The physical metric object remains the lawful orbit; the section is only a local analytic representative used for the Cauchy vector field.

B.6 Exact full-ray quantum descent

The quantum branch does not differentiate the full projective carrier by an unbounded generator. Exact closure is

$$\Pi_H(x) = \Pi_H(y) \implies \Pi_H(\alpha_t x) = \Pi_H(\alpha_t y).$$

On the quantum-eligible subtype the quotient process is additionally a strongly continuous, transition-preserving all-real group. Wigner and Stone are applied only after this exact full-ray process has been constructed. In the complete inhabitant the upstream seed has zero projective component, while

$$\Pi_H(\theta, q, r) = R_\theta q$$

induces the downstream Stone process relationally through the clock coordinate.

B.7 Exact record and compatibility descent

For record and compatibility carriers, no differential is used. Exact closure is

$$P(x) = P(y) \Rightarrow P(\alpha_t x) = P(\alpha_t y).$$

The quotient process is then

$$\beta_t([x]) = [\alpha_t x].$$

This is the universal property of the final quotient, not a smooth-flow assertion on a discrete or categorical carrier.

B.8 Closed versus exchange-extended continuation

Exact quotient descent constructs the closed base process. It does not itself define a counterfactual open prediction. The declared-open construction therefore adds an independently registered upstream incidence ledger and forms a separate enriched projection and process. The lifted closed predictor preserves the incoming ledger; the actual enriched process advances it. Their directed difference is an action morphism, not an equivalence relation.

C Metric projection and Cauchy-intertwining checklist

A metric history enters the dynamic Einstein/source category only after every row is discharged.

Burden	Required certificate
Upstream ancestry	Metric/source presentation obtained from the RPI state by Π_g , never selected as a solved path.
Lawful metric equivalence	Physical profile is the orbit of the complete natural metric/source functional under an independently fixed lawful presentation groupoid; both presentations and compact tests are transported.
Local representative	A genuine local gauge section exists only on the smooth Cauchy-eligible stratum and is used analytically, not as primitive physical identity.
Metric anchors	Four-dimensional Lorentzian carrier, regularity, locality, boundary, gauge, and compatible connection fixed.
Source ancestry	Accepted Module A attachment object where a source is claimed.
Ward status	Zero divergence in closed scope or named J_ν in open scope.
Pointwise response	Tensorial, symmetric, rank two, local, first curvature response, no undeclared extra load.
Smooth Cauchy system	Registered gauge-fixed hyperbolic Einstein–matter reduction and constraint-satisfying initial data.
Seed compatibility	In the local section, $d\mathcal{P}_g^\sigma(V_D) = \mathcal{F}_g \circ \mathcal{P}_g^\sigma$ for the independently defined reduced Cauchy vector field.

Uniqueness	The local Cauchy theorem applies, identifying the projected path with the unique development.
Closed/open distinction	The closed metric process evolves the lawful base orbit. Any declared exchange is a separate enriched ledger coordinate and process, not a second reading of the same quotient process.
No hidden repair	Boundary, source, gauge, or constraint failure terminates the eligible history or enters declared exchange.

Without the lawful-orbit, local-section, smooth Cauchy-system, projected-solution, and uniqueness rows, Module C reports only pointwise Einstein normal-form classification, not dynamical intertwining.

D Quantum projection bridge details

D.1 Complete quantum fibre

The full transition function against every ray determines the current ray. The graph and domain determine the closed generator. Fixed spatial, potential, baseline, and closure ledgers determine the remaining same-scope quantum theorem object. Thus the quantum fibre is proved complete rather than labelled complete.

D.2 Full-ray requirement

A single orbit is insufficient for Wigner's theorem. The saturation certificate requires a quotient map onto the full ray space and a process defined on every ray for the common interval.

D.3 Transition preservation

For every ray pair,

$$\text{TP}(\beta_t[\psi], \beta_t[\phi]) = \text{TP}([\psi], [\phi]).$$

This is simultaneous same-scope transport, not the probability of one evolved state against a fixed apparatus.

D.4 Unitary branch and phase lift

Wigner implementation gives unitary or antiunitary representatives. Strong continuity through the identity excludes the antiunitary branch. A continuous phase choice removes the one-parameter multiplier; a residual constant phase rate is the homogeneous baseline.

D.5 Stone and spatial normal form

Strong continuity gives the self-adjoint generator. The spatial normal form is separate: the fixed scalar nonrelativistic envelope and SSQ_1 give

$$H = a\Delta + M_V,$$

and the accepted inertial/action anchors fix $a = -\hbar^2/(2m)$.

D.6 Forward history and reversible Hilbert group

The Hilbert representative may be mathematically reversible while the admitted RPI history remains clock forward and record fail closed. These are different roles, so no contradiction arises.

E Finite certificate schema

The bounded certificates distinguish base closed profiles from exchange-extended actual profiles. A representative finite object has the shape:

```
{
  "module": "AASC-GRQM-MC-CONT-1.0",
  "certificate_id": "...",
  "clock_representative": "tau-local-01",
  "upstream_seed": {
    "id": "seed-smooth-01",
    "upstream_coordinates": ["theta", "q", "r"],
    "vector_field": ["1", "0", "1"],
    "globally_smooth": true,
    "contains_unbounded_generator": false
  },
  "projection_separation": {
    "metric": {"closed_map": "Pi_metric^0", "actual_map": "tilde_Pi_metric", "
      independently_defined": true},
    "quantum": {"closed_map": "Pi_quantum^0", "actual_map": "tilde_Pi_quantum", "
      independently_defined": true},
    "record": {"closed_map": "Pi_record^0", "actual_map": "tilde_Pi_record", "
      independently_defined": true},
    "compatibility": {"closed_map": "Pi_compatibility^0", "actual_map": "
      tilde_Pi_compatibility", "independently_defined": true}
  },
  "states": [
    {
      "id": "x0",
      "standing": true,
      "failure": false,
      "clock": "0",
      "common_ledger": "0",
      "base_profiles": {
        "metric": "...",
        "quantum": "...",
        "record": "...",
        "compatibility": "..."
      }
    }
  ],
  "exchange_incidence": [
    {
      "id": "inc-01",
      "role_weights": {
        "metric": "1",
```

```

    "quantum": "0",
    "record": "0",
    "compatibility": "0"
  }
},
"morphisms": [
  {
    "id": "f01",
    "source": "x0",
    "target": "x1",
    "duration": "1",
    "path_samples": ["x0", "x01mid", "x1"],
    "common_incidence": "inc-01",
    "common_amount": "1",
    "faces": {
      "metric": {
        "status": "open",
        "base_predicted_target": "...",
        "predicted_enriched_target": {"base": "...", "ledger": "0"},
        "actual_enriched_target": {"base": "...", "ledger": "1"},
        "exchange": "ex-01"
      }
    },
    "cube_vertices": {"0000": "...", "1111": "..."}
  },
  {
    "first": "f01", "second": "f12", "composite": "f02"
  }
],
"accepted_import_hashes": {...}
}

```

The seed registry records only the finite claimed coordinate pattern; it does not prove analytic smoothness. The projection-separation registry prevents the checker from identifying the closed prediction map with the exchange-extended actual map.

For every role, the base predicted target must equal the target state's closed base profile. The predicted enriched endpoint retains the source state's ledger restriction; the actual enriched endpoint carries the target state's ledger restriction. An open exchange amount must equal the common incidence amount multiplied by the registered role weight. Composition must preserve base process targets and add common and role ledger increments. The checker regenerates all sixteen vertices from the predicted and actual enriched endpoints.

The finite checker does not decide local Lipschitz continuity, physical-clock adequacy, lawful metric-orbit finality, Einstein or Schrödinger equations, or standing between registered samples.

F Theorem-use ledger

Source	Used result	Prohibited promotion
AASC kernel	Target, standing, reference, irreversibility, no repair	Does not choose V_D or a clock
RPI	Upstream roles, quotient identity, locality, transport, carrier, mass, vacuum, time, record, and causality	Does not supply one preferred local flow
Coherent trajectory theory	Prefix/composition closure and irrecoverability	Does not generate licensed transitions
Physical clock quotient	Certified local clock class and clock-use profile	Does not select dynamics or identify durations across nonlinear reparametrizations
Connection theorem	Quotient-stable transport comparison	Does not supply the Module C seed
Persistence realization	Persistence class from an existing irreversible lineage	Does not create the lineage
Module A	Attachment-safe source, Ward/exchange, finite composite closure	Does not supply an upstream history
Module B	Stable scalar active/inertial matching in its exact envelope	Does not supply an upstream history
Einstein paper	Minimal metric normal form	Does not prove every source Cauchy problem
Einstein Cauchy theorem	Local uniqueness for a fixed smooth hyperbolic system	Does not authorize the upstream seed or arbitrary distributional sources
Schrödinger paper	Wigner–Stone and spatial normal form	Does not turn one ray path into a full-ray flow
Record paper	Final record quotient and fixation	Does not select an outcome without a ledger
Entanglement paper	Final compatibility-not-transport classification	Does not supply source transfer or signaling

G Claim-status matrix

Claim	Status	Boundary
Local upstream flow from certified seed	Proved here	Seed is explicit tensor content, not kernel generated
Fixed-clock Moore category	Proved here	One declared local clock representative

Clock-presentation equivalence	Proved here	Equivalence between separate duration categories
No deferred history repair	Proved/imported	Same lineage and fixed target
Metric-fibre completeness	Proved here	Lawful groupoid orbit of full natural metric/source functional
Quantum-fibre completeness	Proved here	Fixed full-ray, operator-domain, and spatial/source data
Record-fibre completeness	Imported/finalized here	Declared record-use scope
Compatibility-fibre completeness	Imported/finalized here	Declared compatibility witness scope
Metric Cauchy intertwining	Proved here	Smooth registered hyperbolic subtype only
Quantum intertwining	Proved here	Exact strongly continuous full-ray scalar subtype
Record and compatibility descent	Proved here	Exact fibre preservation, no tangent assumption
Distinct closed and enriched role processes	Proved here	Split exchange extension
Directed exchange action and cocycle	Proved here	Declared common incidence monoid
Sixteen-vertex continuation four-cube	Proved here	Product action category
Complete strict nonidentity inhabitant	Proved here	Smooth clock-translation RPI packet
Complete composable open inhabitant	Proved here	Separate closed prediction and actual metric ledger
Nonzero interacting quantum gravity	Not claimed	Later richer-scope target
Global clock or global flow completeness	Not claimed	Local clock/use envelope
Capstone unique orbit	Not claimed	Module D

H Standalone hostile-referee checklist

A hostile but endpoint-faithful bounded Cycle-3 review should answer each question with *correct*, *incorrect*, or *unresolved*.

1. Is the kernel dependency order preserved?
2. Is the seed independently formed on the RPI quotient before projection?
3. Is one clock representative fixed in the Moore category, with other representatives related only by a duration-transport functor?

4. Does local-flow existence follow from an actual regularity theorem or an explicit smooth inhabitant?
5. Are standing preservation, nonidentity, and no deferred repair proved?
6. Is the metric profile a lawful orbit with both tests and source data transported, rather than equality of fixed representative pairings?
7. Is any gauge section used only as a local analytic representative of the lawful orbit?
8. Are the quantum, record, and compatibility fibres independently final in their fixed targets?
9. Are smooth metric descent, exact full-ray quantum descent, and exact record/compatibility descent correctly distinguished?
10. Is the metric process genuinely identified with a unique local Einstein–Module-A Cauchy development on the exact eligible subtype?
11. Does the quantum theorem use a full-ray transition-preserving strongly continuous group, with the unbounded generator introduced only downstream?
12. Are the closed base projection/process and actual exchange-extended projection/process distinct maps and theorem objects?
13. Is the common incidence ledger independently registered, with role restrictions and a correctly oriented source-target-compatible cocycle?
14. Is the continuation object a genuine sixteen-vertex four-cube with commuting faces, monotone paths, and compositions?
15. Does the closed packet use a globally smooth upstream seed and nevertheless provide full-ray quantum saturation by a clock-conditioned projection?
16. Does the open packet contain two composable histories verifying a nonzero exchange cocycle without changing the closed base role?
17. Are examples reported only at their actual strict or declared-open scope?
18. Does the software claim remain bounded to the finite registry it checks?

A review may invalidate the module by locating a failed implication at one of these coordinates. It may not demand Module D uniqueness, nonzero interacting quantum gravity, or global clock existence.

I Symbol index

Symbol	Meaning
K_D	AASC kernel (Adm, Stand, Ref, Irr)
$I_D, \bar{I}_D$	realized physical interior and lawful physical quotient
U_D	regular local continuation chart
$[\Theta]_{\text{Clk}, U}, \tau$	certified local clock class and fixed representative
V_D	upstream continuation seed

$\alpha_{t,s}$	local upstream flow
$\mathbf{Cont}_D^{\tau, \text{raw}}, \mathbf{Cont}_D^{\tau}$	raw and fixed-clock lawful Moore history categories
$\mathfrak{T}_f$	duration-transport functor between clock presentations
$\mathcal{G}_g$	lawful metric/source presentation groupoid
OrbProf_g	orbit of the complete natural metric/source functional
Y_X	final closed role carrier
Π_X^0	closed base role projection
β^X	unique closed descended role process
$\tilde{Y}_X = Y_X \times \mathcal{J}_X$	exchange-extended role carrier
$\tilde{\Pi}_X$	actual exchange-extended role projection
$\hat{\beta}_X^0$	zero-new-exchange enriched prediction
$\tilde{\beta}_X$	actual enriched role process
$\mathbf{Ein}_D, \mathbf{Sch}_D$	Einstein and Schrödinger process categories
$\mathbf{Rec}_D, \mathbf{Comp}_D$	record and compatibility process categories
$\mathcal{J}_D$	common conical exchange-incidence monoid
J_D	common history incidence
ρ_X	role restriction homomorphism
$\mathcal{E}_X(f; j)$	directed role exchange action morphism
$\mathcal{K}_{f,j}$	sixteen-vertex continuation four-cube
$\hat{Q}_{\text{src}}$	accepted Module A structured source quotient
$\text{SameScopeMetric}_1, \text{SSQ}_1$	fixed metric and quantum first-response predicates
$\mathcal{A}_{\text{Ref}} / \simeq_{\text{MR}}, \mathcal{Q}_{\text{comp}}$	final record and compatibility quotient carriers

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